National Institute Of Business Management

Diploma In Software Engineering

GADSE251F Batch

Web-Based Tourism Services Marketplace Platform

Final Project Report

("Ceylon Tour Mate.lk")

Submitted by:
Name: K.K.Naveen Rashmika Devinda
Registration No: GADSE251-F-018

Submission Date: 2026-07-27

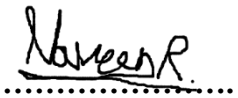
National Institute of Business Management (NIBM) – Galle Campus

No. 132, Samudhra Mawatha (Sea Street), Galle 80000, Sri Lanka

Declaration

I, Naveen Rashmika Devinda, hereby declare that this final project report submitted for DSE final project is an original work done by me under the guidance of Ms.Ashanthi Kurakulasuriya ,also declare that this report record has not been submitted to any other institute.

Signature:



Naveen Rashmika Devinda

GADSE251F-018

Date:

2026.07.27

Supervised by:

Ms. Ashanthi Kurakulasuriya

(Consultant/Lecturer - National Institute of Business Management)

Abstract

Tourism industry of Sri Lanka mainly relies on multiple platforms to connect tourists with local tour guides, hotels, shops and experience providers. However this causes problems like inefficient bookings, limited chances for small businesses and lack of a centralized platform.CeylonTourMate.lk is to fill that gap by providing verified local service providers to tourists on a web based centralized platform.

Its developed using HTML, PHP, CSS and MySql with OOP architecture with all the classes.it handles role based login, listing management, dashboards for users, messaging system, report generations and a online payment method.

This report includes the full software development process of CeylonTourMate.lk from scratch with incorporating with supervisor feedbacks. This project helps to solve a problem using a practical solution and improves digital access to Sri Lankas tourism industry.

Contents

Table of figures v

Table of tables vii

Chapter 1: Introduction 1

1.1 Introduction of the Organization 1

1.2 Organization Structure 1

1.3 Current Operations in Organization 2

1.4 Users and Responsibilities in Organization 2

1.5 Problem Definition 2

1.6 Project Objectives 3

1.7 Proposed Solution 3

1.8 Chapter Summery 3

Chapter 2: Methodology 4

2.1 Introduction 4

2.2 Data Collection Methods 4

2.3 Software Process Model 4

2.4 Software Development Tools 5

2.5 Testing Strategies 5

2.6 Implementation Plan 5

2.7 Chapter Summary 6

Chapter 3: Analysis 7

3.1 Introduction 7

3.2 UML Diagrams 7

3.2.1 Use Case Diagram of Current System 7

3.2.2 Use case Diagram of Proposed System 8

3.2.3 Class Diagram of Proposed System 9

3.2.4 Sequence Diagrams of Proposed System 10

3.3 ER Diagram of Proposed System 13

3.4 Chapter Summery 14

Chapter 4: Solution Design 15
4.1 Introduction 15
4.2 Interface Design 15
4.3 Database Design 26
4.4 Report Layout Design 31
Chapter 5: Implementation 34
5.1 Introduction 34
5.2 System architecture 34
5.3 Main Coding Areas 35
5.4 Admin Report Implementation 40
Chapter 6: Testing 41
Chapter 7: Conclusion 42
7.1 Achievements 42
7.2 Limitations 42
7.3 Future Developments 42
7.4 Final Conclusion 42
References 43

Table of figures

Figure 1 Organization Structure 1
Figure 2 Use Case Diagram of Proposed System 8
Figure 3 Class Diagram of Proposed System 9
Figure 4 Sequence Diagram for Admin Approve/Reject Listings 10
Figure 5 Sequence Diagram for Admin Approve/Reject Registrations 11
Figure 6 Sequence Diagram for Tourist Booking a Listing 12
Figure 7 ER Diagram of Proposed System 13
Figure 8 Interface 1 15
Figure 9 Interface 2 16
Figure 10 Interface 3 16
Figure 11 Interface 4 17
Figure 12 Interface 5 17
Figure 13 Interface 6 18
Figure 14 Interface 7 18
Figure 15 Interface 9 19
Figure 16 Interface 8 19
Figure 17 Interface 10 20
Figure 18 Interface 11 20
Figure 19 Interface 12 21
Figure 20 Interface 13 21
Figure 21 Interface 14 22
Figure 22 Interface 15 22
Figure 23 Interface 15 23
Figure 24 Interface 16 23
Figure 25 Interface 18 24
Figure 26 Interface 17 24
Figure 27 Interface 19 25
Figure 28 Interface 20 25
Figure 29 report layout 1 31
Figure 30 report layout 2 32
Figure 31 report layout 3 32
Figure 32 report layout 4 33
Figure 33 report layout 5 33
Figure 34 Main classes 34
Figure 35 Session class 35
Figure 36 Welcome message 35

Figure 37folder structure 35
Figure 38githubcommits 40
Figure 39commits over time 40
Figure 40 mPDF 40

List of tables

Table 1 User Responsibilities 2
Table 2 Software tools 5
Table3:Users 26
Table4:listings 26
Table5:listing_images 27
Table7:rooms 27
Table6:hotel_listings 27
Table10:tourguide_listings 28
Table9:experience_listings 28
Table8:shop_listings 28
Table13:booking_rooms 29
Table12:bookings 29
Table11:tourguide_licenses 29
Table14:payments 30
Table15:reviews 30
Table16:messages 30
Table17:notification 31
Table18:commission 31
Table19Test cases 41

Chapter 1: Introduction

1.1 Introduction of the Organization

Tourism has become one of the fastest growing industries in Sri Lanka. Every year it keeps growing, according to the Sri Lankan Tourism Development Authority about 2.3 million tourists visited Sri Lanka in last year exceeding previous records.

With this growth a lot of opportunities has created for Sri Lankans including hotel owners, tour guides, shop owners and local experience providers. They offer tourism related services as their main income.

For tourists getting these information become bit hard and tricky due to spread of these info on many platforms.

So the proposed solution for this problem is a centralized platform named "**CeylonTourMate.lk**" which can connect tourists and service providers across Sri Lanka.

1.2 Organization Structure

These are the main roles in the platform

- Admin
- Tourist
- Tour Guide
- Hotel Owner
- Shop Owner
- Experience Provider

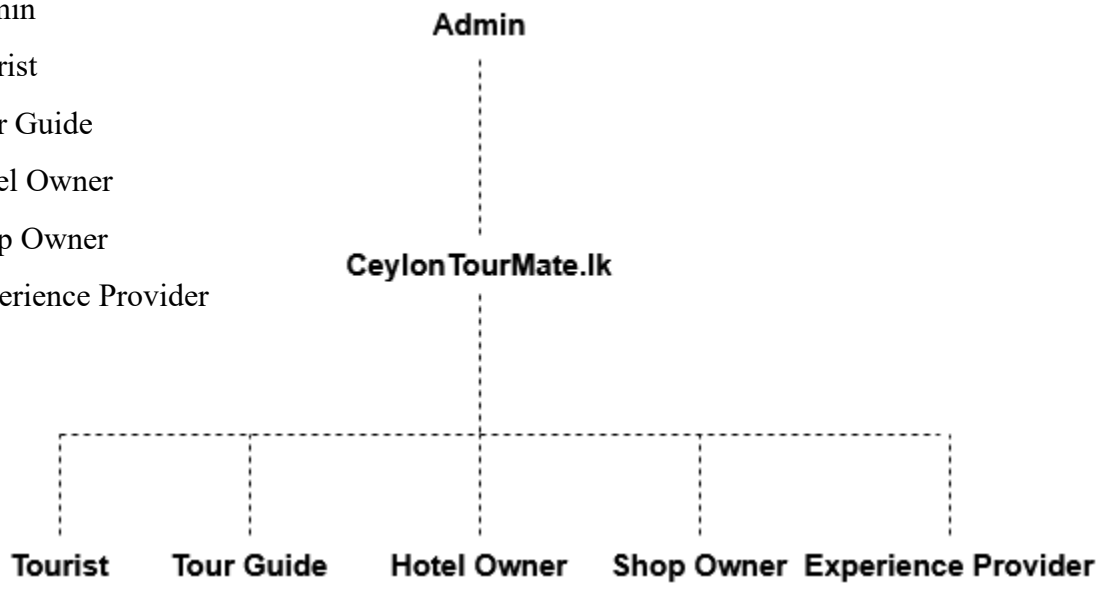


Figure 1 Organization Structure

1.3 Current Operations in Organization

Currently information about tour guides, hotels, traditional shops and local experiences in Sri Lanka are spread across multiple platforms like social media, third party booking sites.

Also there isn't any centralized platform for tourists to get these information from verified local service providers as well as local businesses to list their listings for international audience.

1.4 Users and Responsibilities in Organization

Role	Responsibilities
Admin	Approve/reject partner registrations and listings, manage users, set commission and listing fee rates, monitor bookings, generate reports
Tourist	Search/browse listings, make bookings, send messages to partners, write reviews, complete payments
Tour Guide	Manage listings, set availability, handle bookings, communicate with tourists, view earnings
Hotel Owner	Manage listings, manage rooms availability, handle bookings, communicate with tourists, view earnings
Shop Owner	Manage listings, set operating hours, pay monthly listing fees, reply to messages
Experience Provider	Manage listings, set availability, handle bookings, communicate with tourists, view earnings

Table 1 User Responsibilities

1.5 Problem Definition

At the moment tourists who visit Sri Lanka face several challenges.

- **Overpriced and unlicensed guides**- Tourists specially first time visitors sometimes meet overpriced or unlicensed guides who try to overcharge or scam them, which is going to be a bad experience for them.
- **Hidden places**- Sri Lanka has many small villages and attraction places that are not included on current tourism platforms, leaving tourist not knowing information about them.

- **Language barrier**- this is the most common issue when it comes to traveling. Not every Sri Lankan can speak English as well as every tourist can speak Sinhala and Tamil. This can be bit challenging when getting a service.
- **Finding services**- Common services like transportation, places to stay and experiences are now spread over many platforms and it can be difficult for both tourists and people who are willing to provide these services.

1.6 Project Objectives

My project objectives are,

- To develop and design a centralized platform to connect tourists with verified local Hotel/Villa owners, Tour guides, Shop owners and Experience providers in Sri Lanka.
- To give small local business a chance to list their listing to the international audience.
- To minimize the risk of tourists getting scammed and making it easy to get those services through one trusted platform.
- To earn from the platform by getting a commission from bookings and listing fee from shop listings.

1.7 Proposed Solution

CeylonTourMate.lk is a centralized platform that built for the Sri Lankas tourism industry that provides all kinds of trustworthy information about local services to tourists.

1.8 Chapter Summery

In this chapter I introduced my project, CeylonTourMate.lk along with the problem of current industry. With the growing of tourism industry, for problems faced by tourists and locals, and as a solution for that I proposed a centralized platform to connect them all.

Chapter 2: Methodology

2.1 Introduction

In this chapter im going to describe how I gather information for my project and approaches I used to develop it. This covers data collection, software development model, tools I used, technologies and testing methods.

2.2 Data Collection Methods

Since my project is a self created project rather than a sponsored one, I had to do following methods to collect data.

- Current market analysis – I did reviews on existing tourism platforms, gathered information from Sri Lankan Tourism Development Authority, viewed attractional places on google maps, and read articles to understand the current gap in the industry.
- Personal knowledge and observation – I found current tourists problems on social media like unlicensed guides, language barrier, etc.. and had a common knowledge of other challenges to tourists.
- Supervisor consultations – since the project started I had meetings with my supervisor Ms.Ashanthi Kurakulasuriya got feedbacks and recommendations throughout development of the project.

2.3 Software Process Model

To develop CeylonTourMate.lk I used the **Waterfall Method,** which is a clear, structured way to develop, since I had a clear idea of my project and its requirements.

The stages of the waterfall method are,

1. **Requirements Analysis** – in this stage I gathered all the requirements for the system and made the SRS document.
2. **System Design** – in this stage I designed all the diagrams.
3. **Coding** – in this stage I started to coding the platform.
4. **Testing** – in this stage I tested my system.
5. **Deployment** – in this stage I hosted my web on a live server.

2.4 Software Development Tools

Category	Tool
Frontend	Notepad++(HTML,CSS,JS)
Backend	Notepad++(PHP)
Database	MySQL(PhpMyAdmin)
Local Server	XAMPP
Code Editor	Notepad++
Version Control	GitHub
Documentation	Microsoft Word
Diagramming	Draw.io
Report Generation	mPDF

Table 2 Software tools

2.5 Testing Strategies

I applied following testing strategies to my project.

- Unit Testing – using this I tested individual php files to confirm their work.
- Functional Testing – tested each functions in the system to confirm their work.
- Validation Testing – tested all form inputs and validations.
- Regression Testing – after found some bugs I fixed them and retested them.

2.6 Implementation Plan

My implementation plan was as following order.

1. Gathering all the requirements and making the SRS document
2. Use case diagram drawing
3. ER diagram designing
4. Class diagram designing
5. Made the database in MYSQL as the ER diagram
6. Coded all the classes like in class diagram
7. Made the Main interface home.html
8. Admin panel coding with all other interfaces
9. Tourist panel coding with all other interfaces

10. Tour guide panel coding with all other interfaces
11. Shop Owner panel coding with all other interfaces
12. Hotel Owner panel coding with all other interfaces
13. Experience Provide panel coding with all other interfaces
14. Testing bug fixing
15. Final report documentation and presentation making

2.7 Chapter Summary

In this chapter I described how I gathered my data to develop my project, software development model, tools I used for development and documentation, testing strategies, and overall methodology of my project.

Chapter 3: Analysis

3.1 Introduction

In this chapter im going to present my analysis I did for CeylonTourMate.lk, including usecase diagram, class diagram, sequence diagram and ER diagram. These diagrams show how the platform works and its database works.

3.2 UML Diagrams

3.2.1 Use Case Diagram of Current System

Since I made this project to fill a gap in the industry, where services are currently spread across multiple platforms, there is not any current system.

3.2.2 Use case Diagram of Proposed System

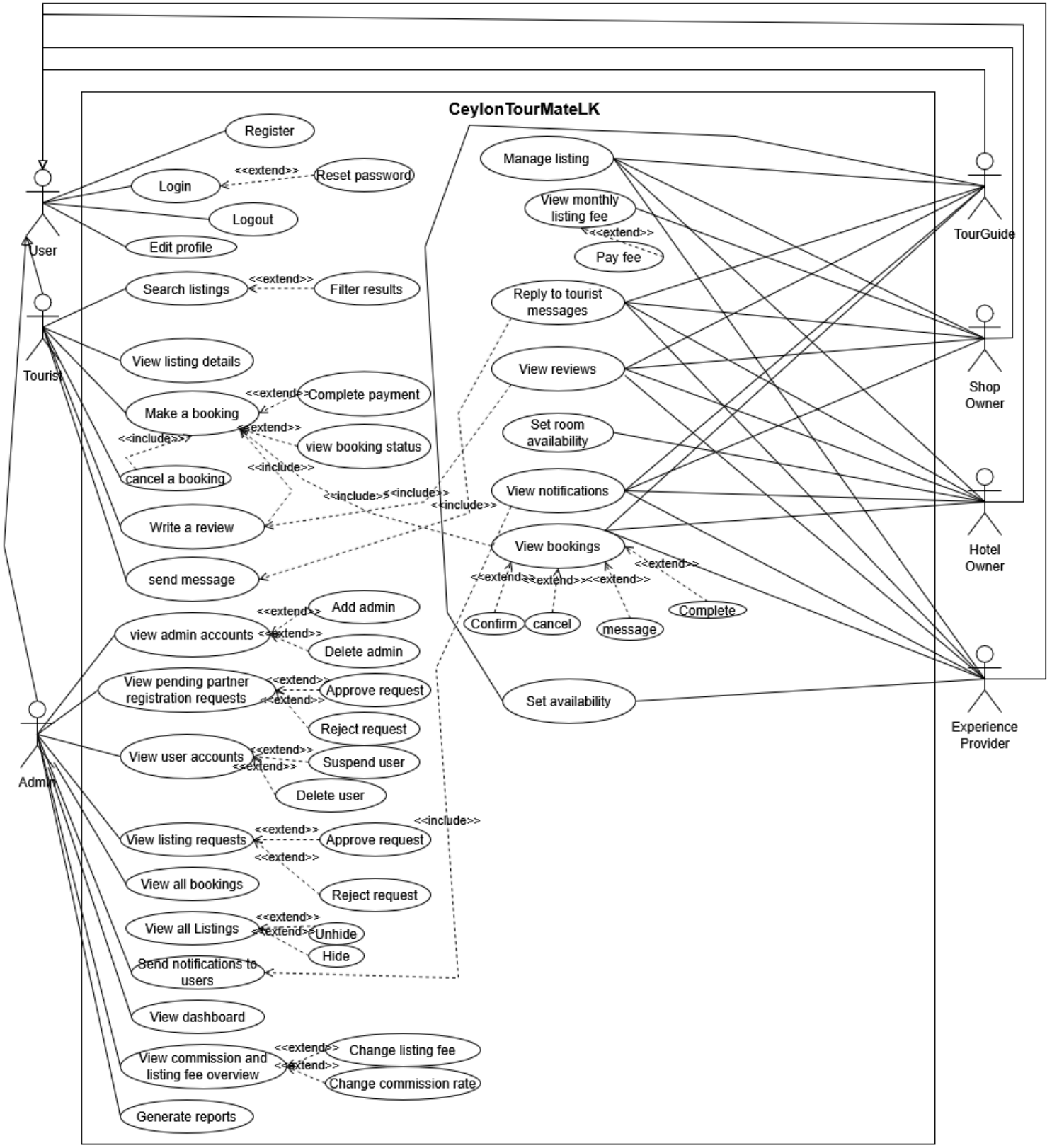


Figure 2 Use Case Diagram of Proposed System

3.2.3 Class Diagram of Proposed System

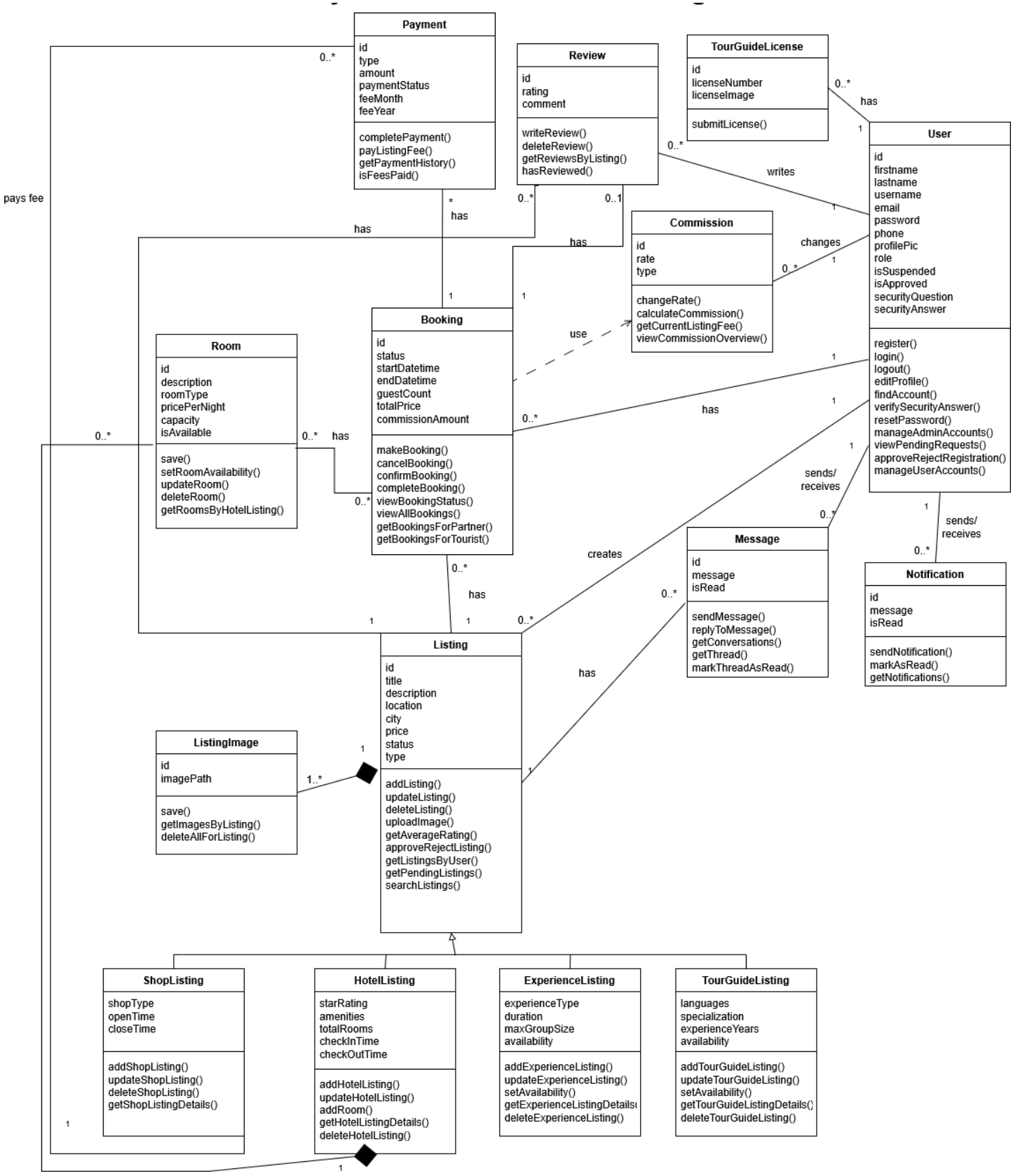


Figure 3 Class Diagram of Proposed System

3.2.4 Sequence Diagrams of Proposed System

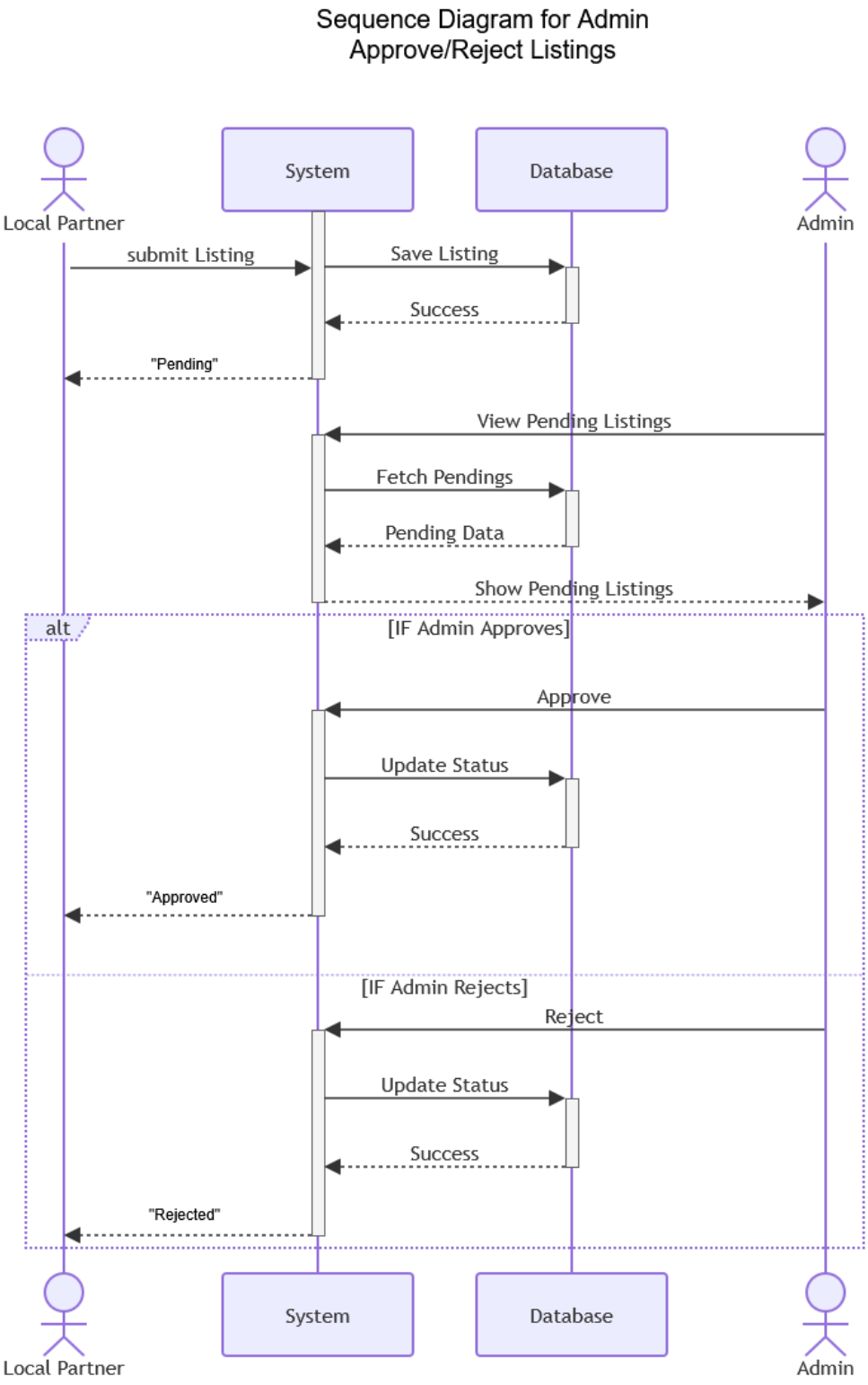


Figure 4 Sequence Diagram for Admin Approve/Reject Listings

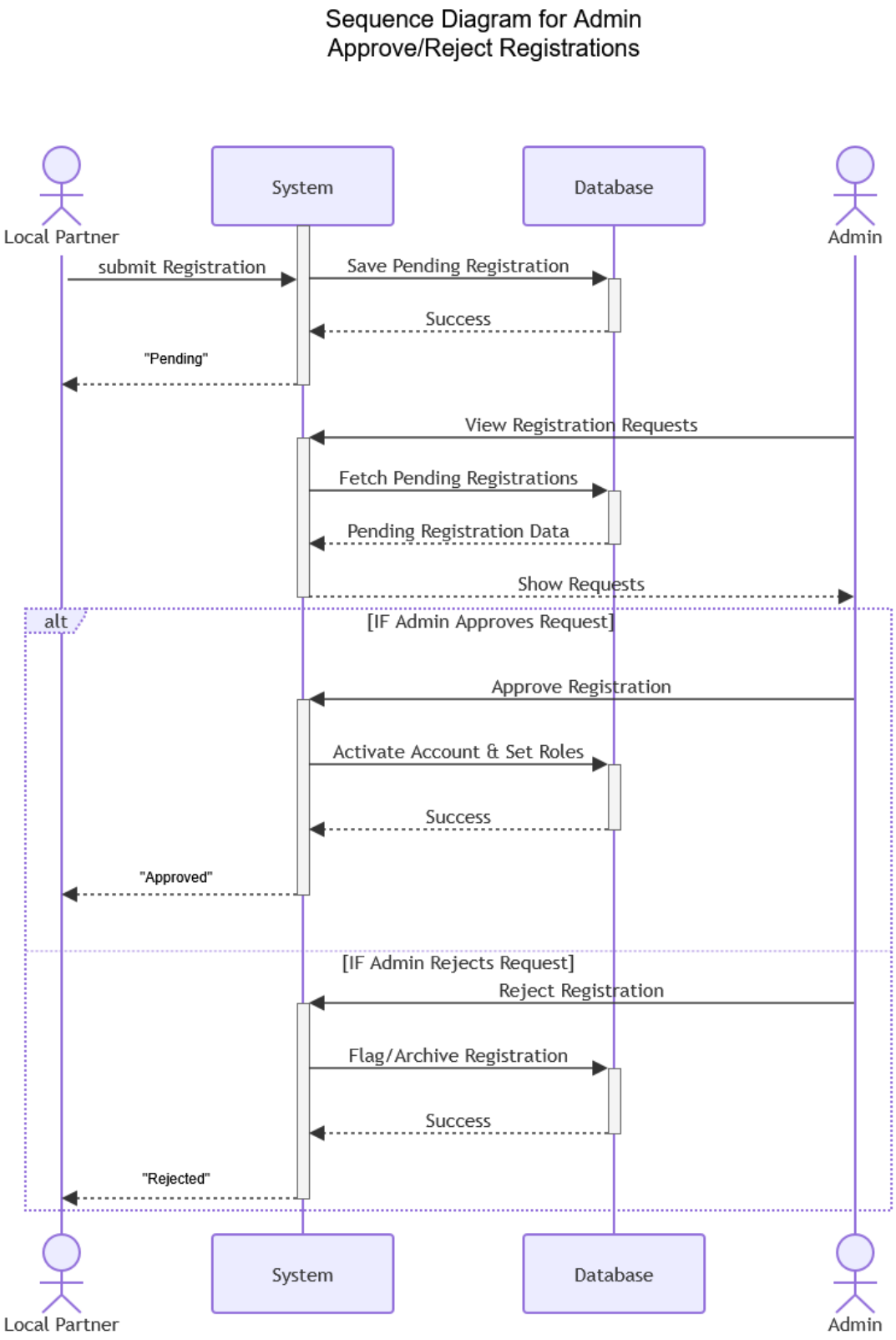


Figure 5 Sequence Diagram for Admin Approve/Reject Registrations

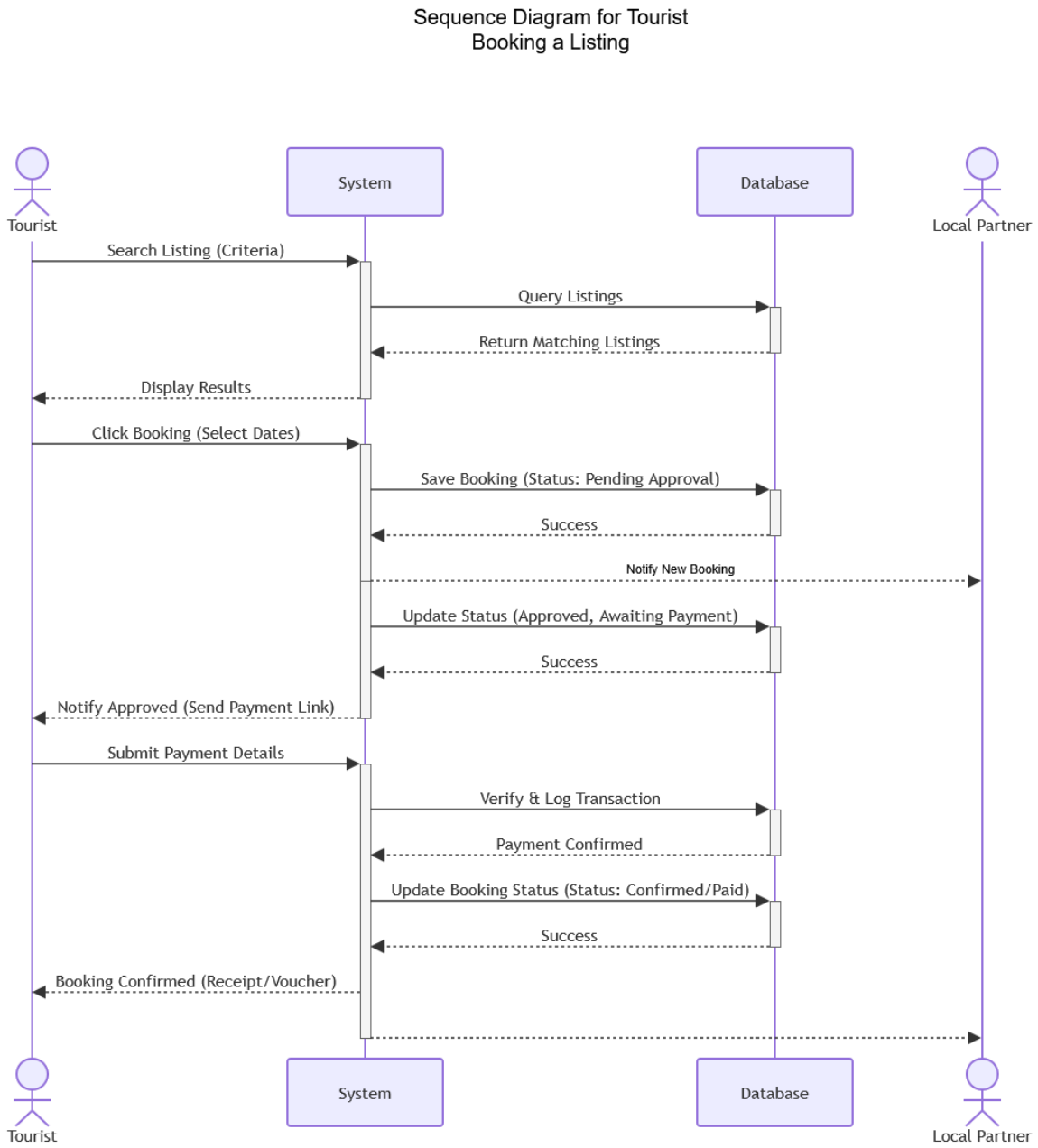


Figure 6 Sequence Diagram for Tourist Booking a Listing

3.3 ER Diagram of Proposed System

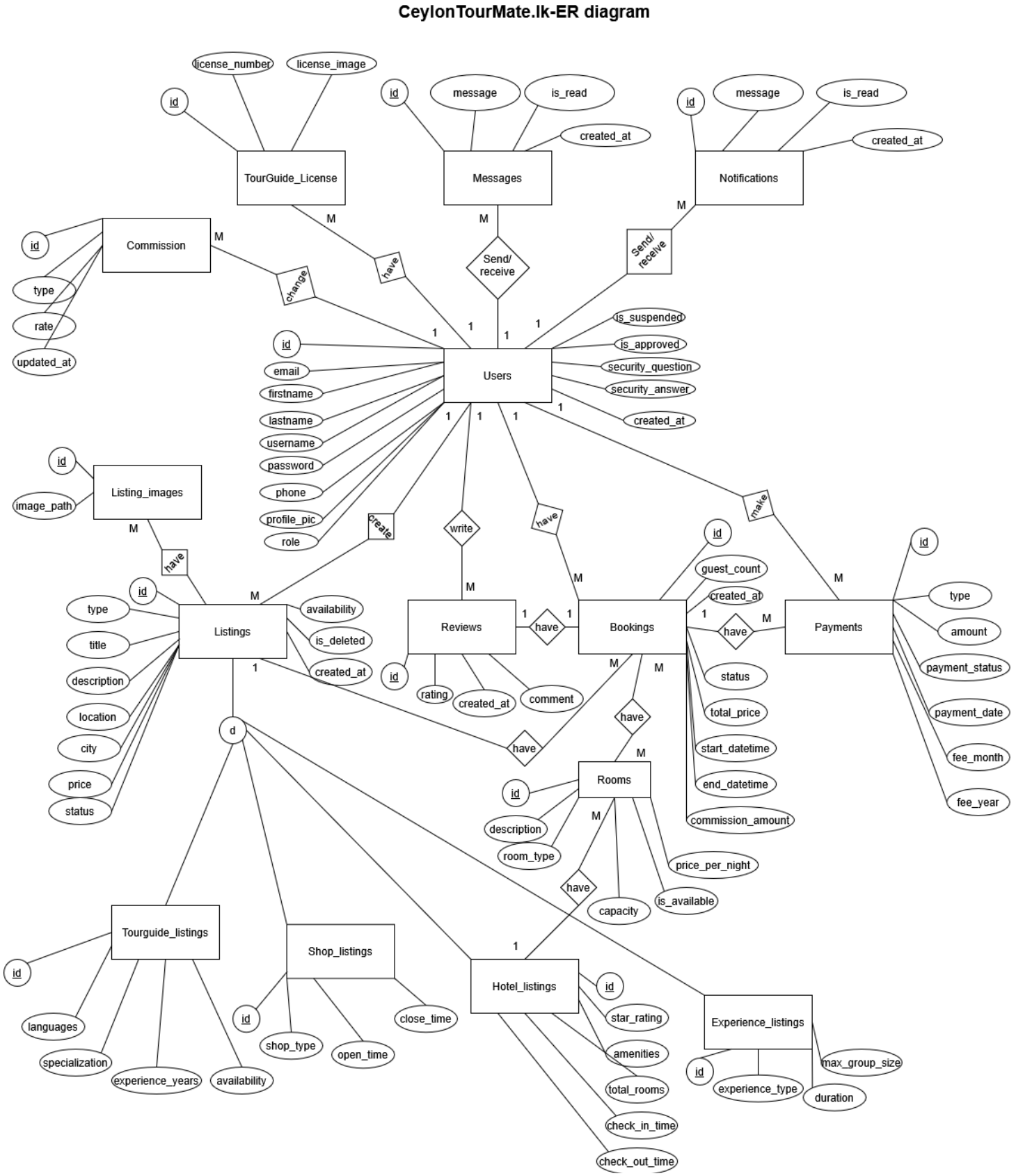


Figure 7 ER Diagram of Proposed System

3.4 Chapter Summery

In this chapter I presented all of my diagrams including usecase, class, sequence and ER. These represent the analysis and modeling stage my CeylonTourMatelk.Use case diagram describes the system actors and their interactions, class diagram describes system's OOP classes, sequence diagram describes the flow of each use case and finally ER diagram describes the database design.

Chapter 4: Solution Design

4.1 Introduction

In this chapter im going to present the design of the proposed solution for CeylonTourMateLK, which include all interface designs, database design, and the report layouts.

4.2 Interface Design

- INT-01 Home Page – Landing page of the platform

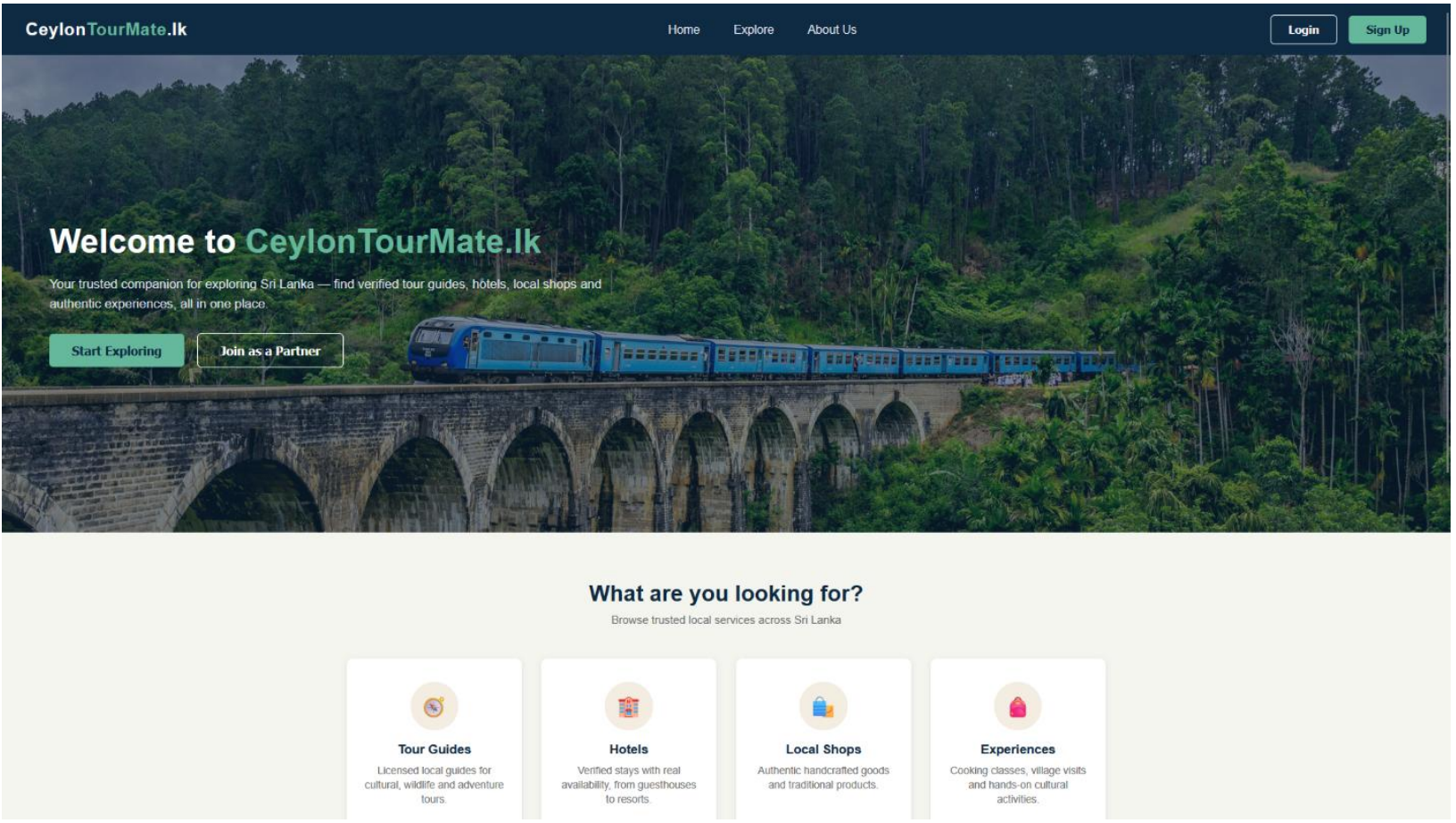


Figure 8 Interface 1

- INT-02 Registration Page – allow all users to register in the platforms, for local partners admin approval required.

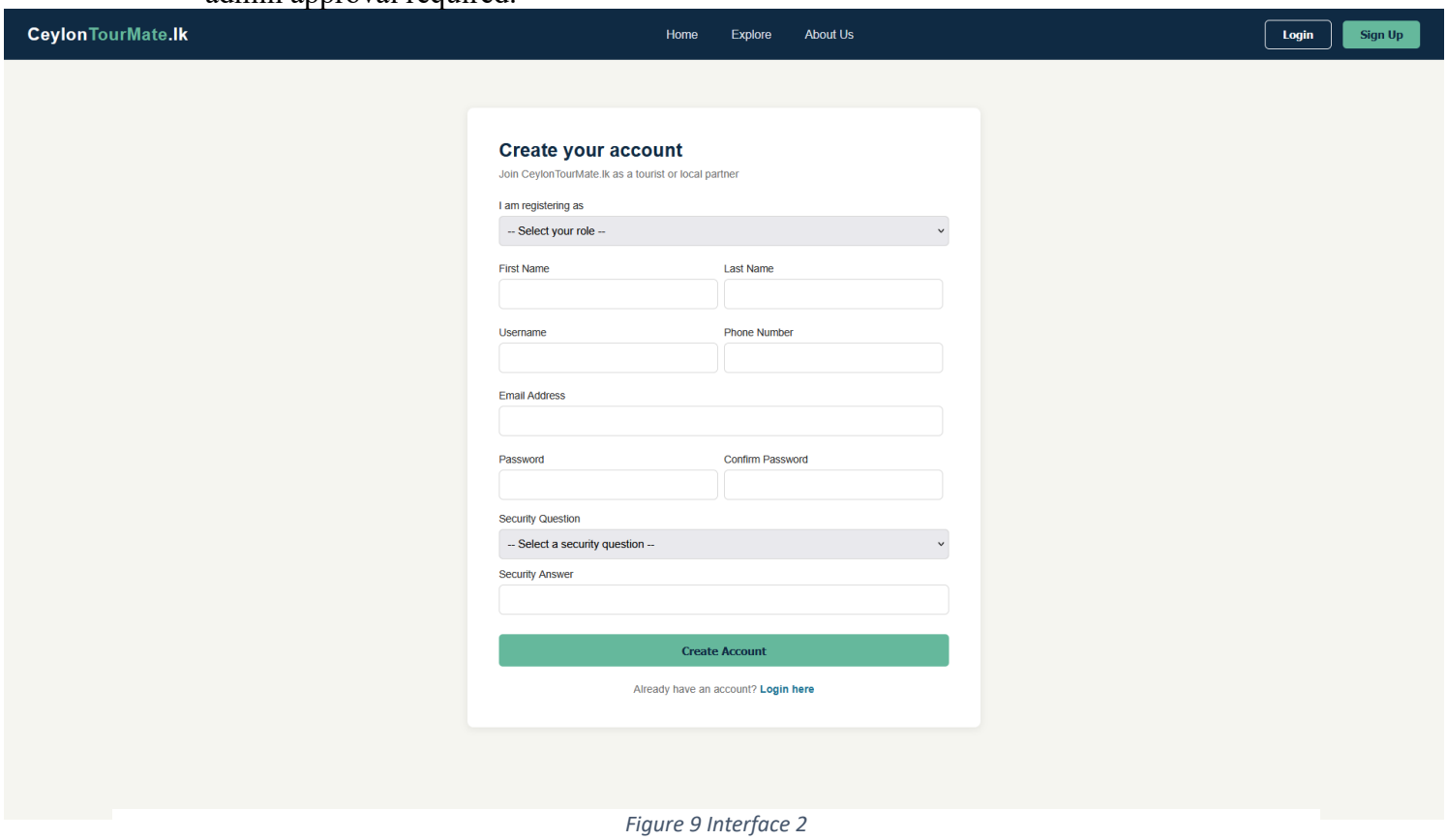


Figure 9 Interface 2

- INT-03 Login page – allows users to login to the system, role based login.

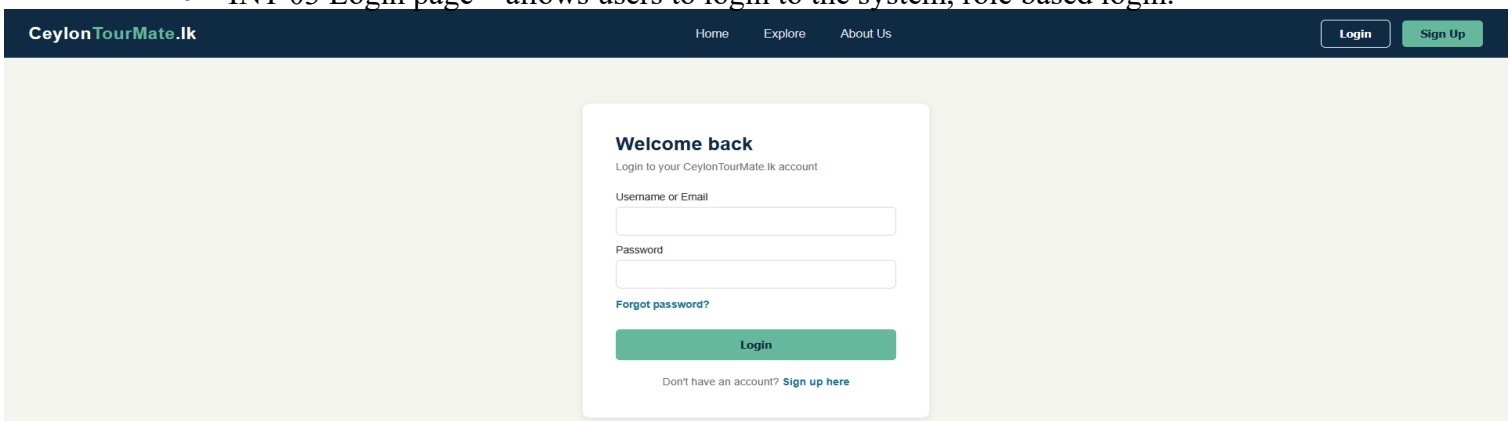


Figure 10 Interface 3

- INT-04 Admin dashboard – where admin can see his dashboard and other controls.

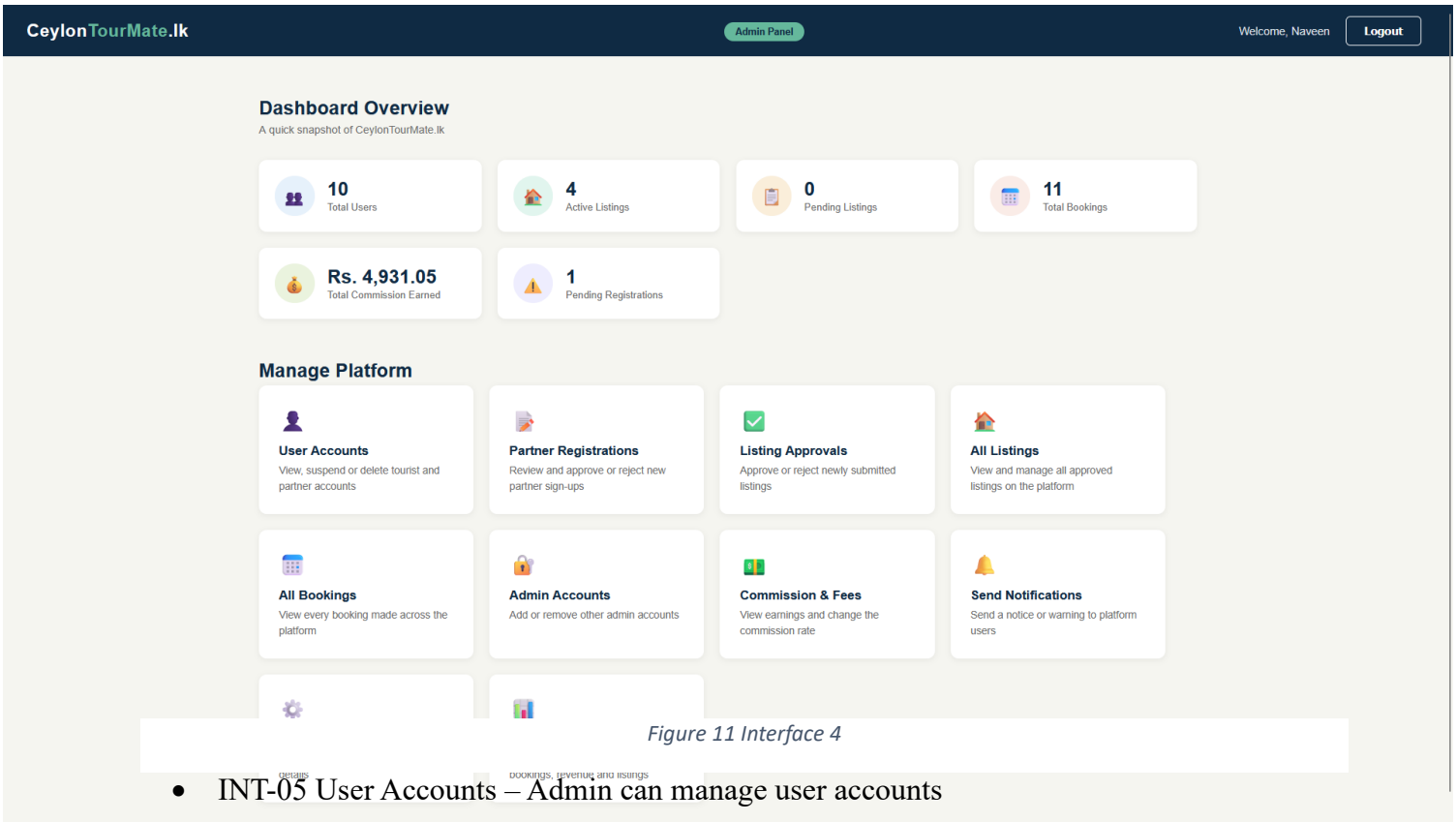


Figure 11 Interface 4

- INT-05 User Accounts – Admin can manage user accounts

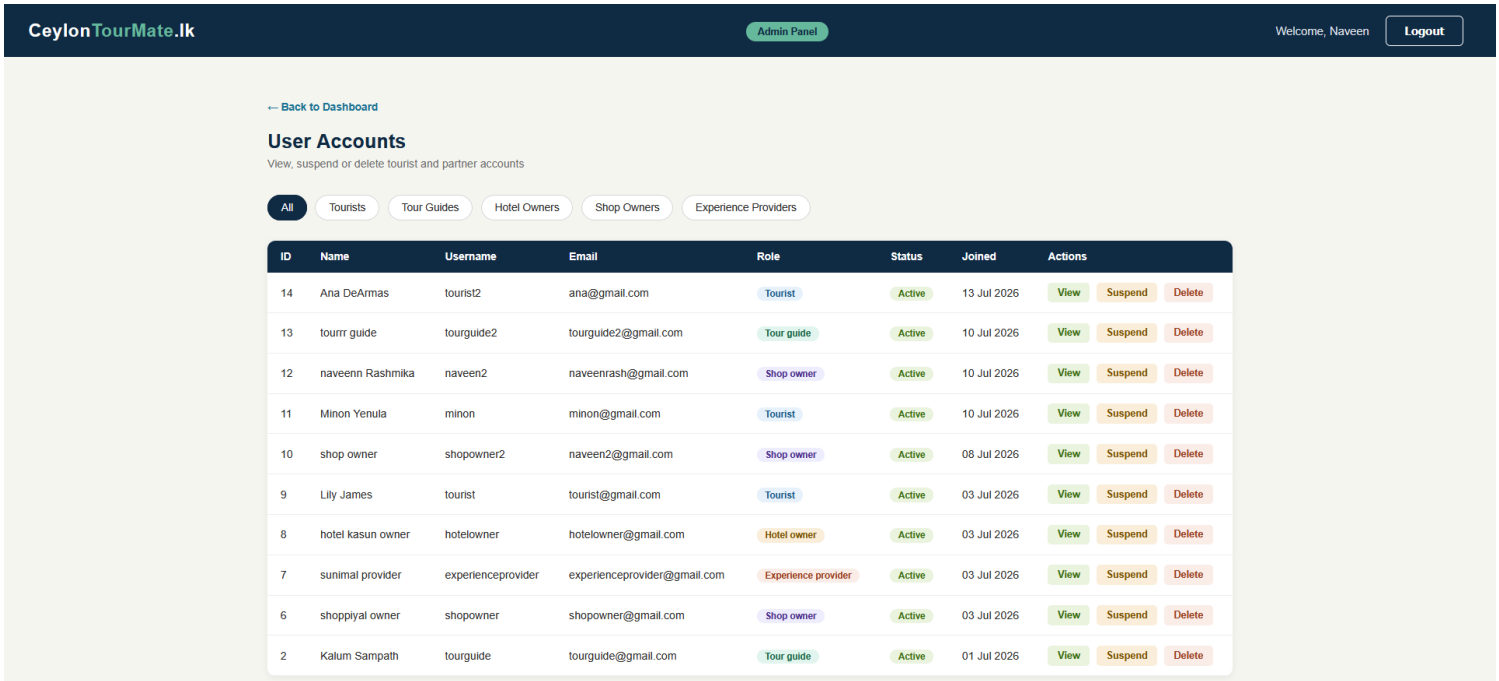


Figure 12 Interface 5

- INT-06 Parter registrations – admin can manage parter registrations

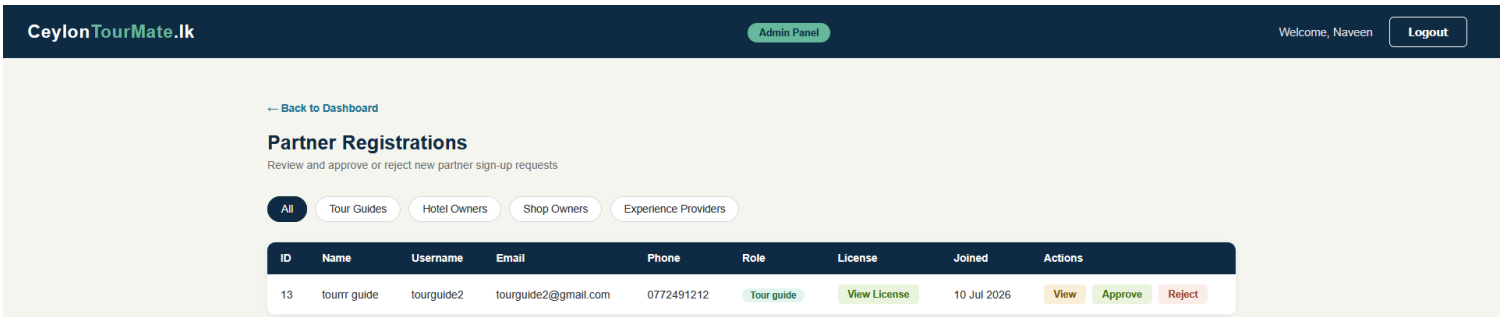


Figure 13 Interface 6

- INT -07 Listing approvals – admin can manage partner listings

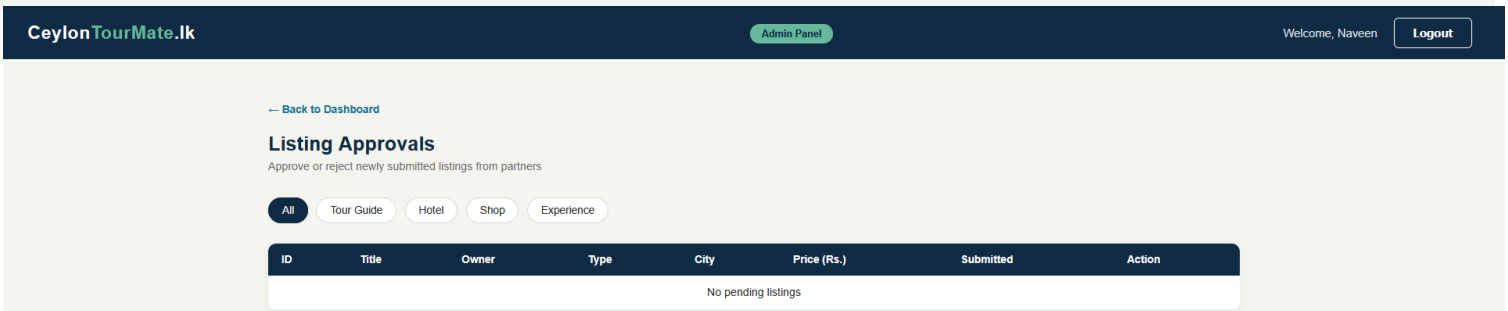


Figure 14 Interface 7

- INT -08 All listings – admin can manage partner listings

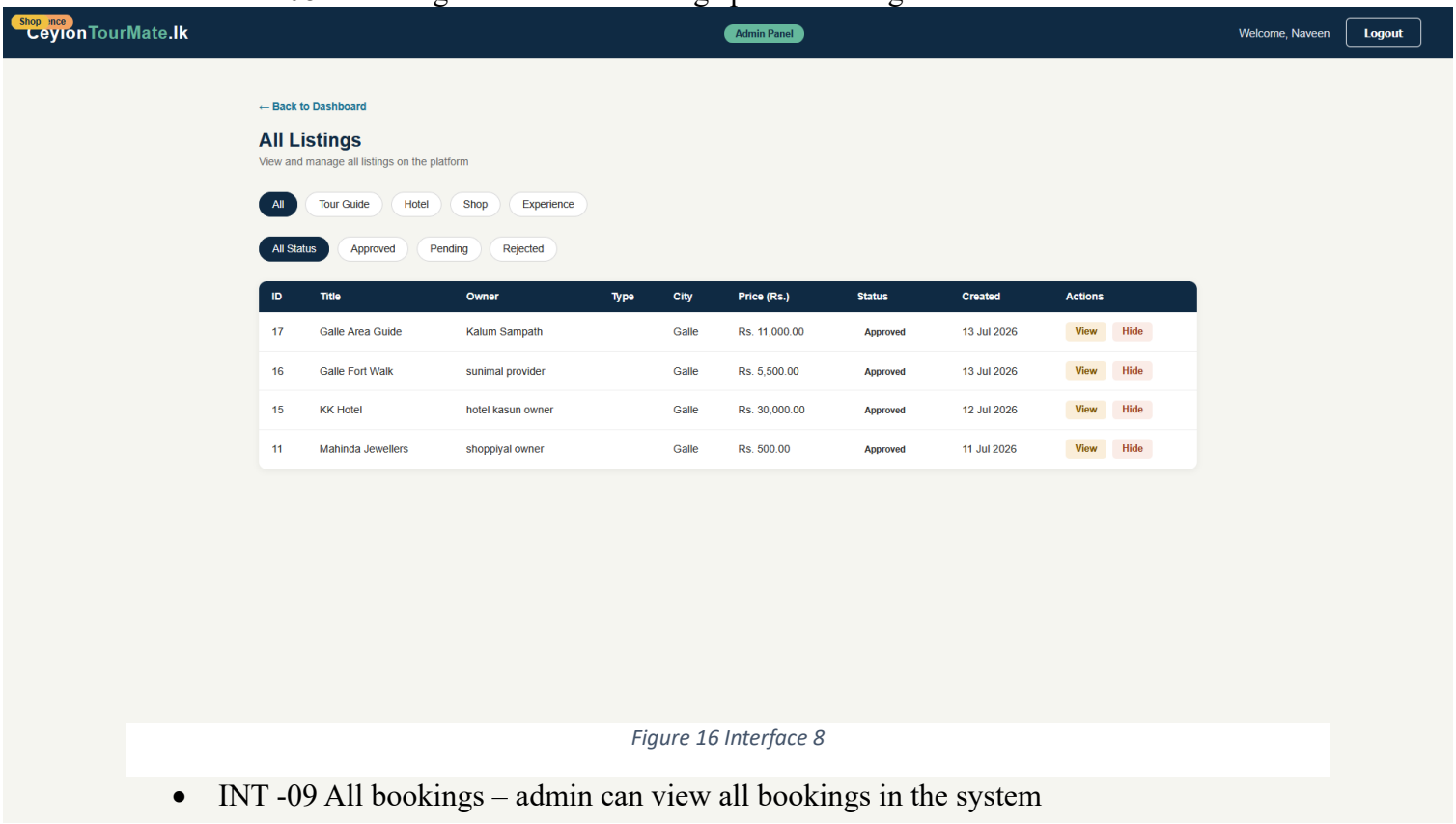


Figure 16 Interface 8

- INT -09 All bookings – admin can view all bookings in the system

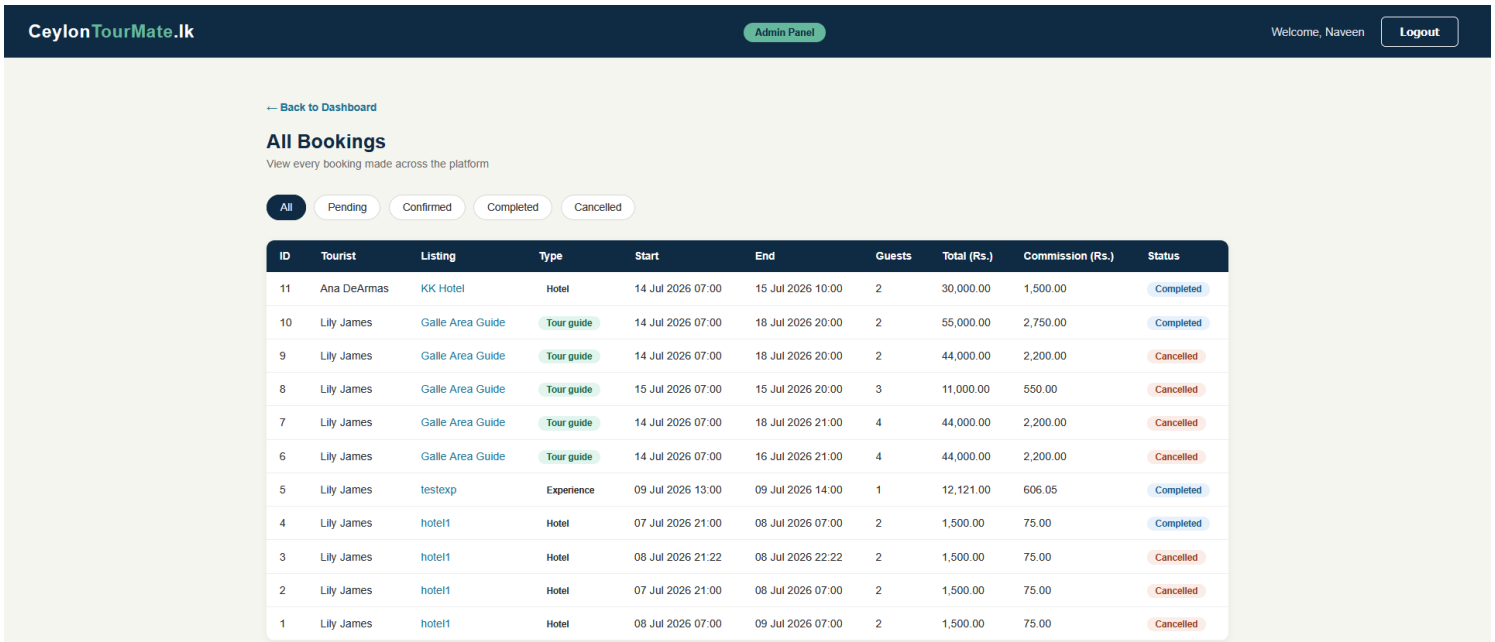


Figure 15 Interface 9

- INT -10 Admin accounts -admin can manage admin accounts

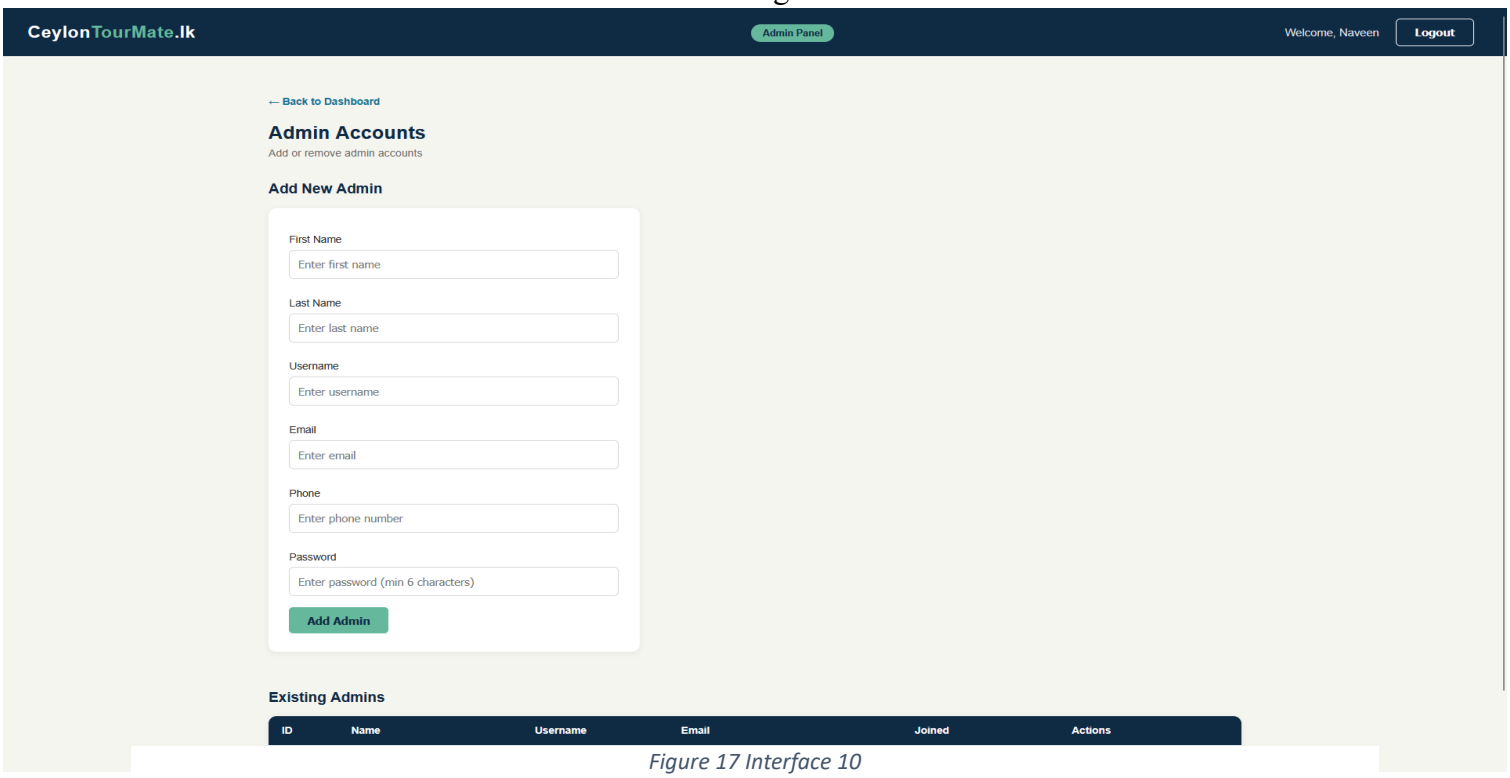


Figure 17 Interface 10

- INT -11 Commission fee overview – admin can view current com rates and manage them

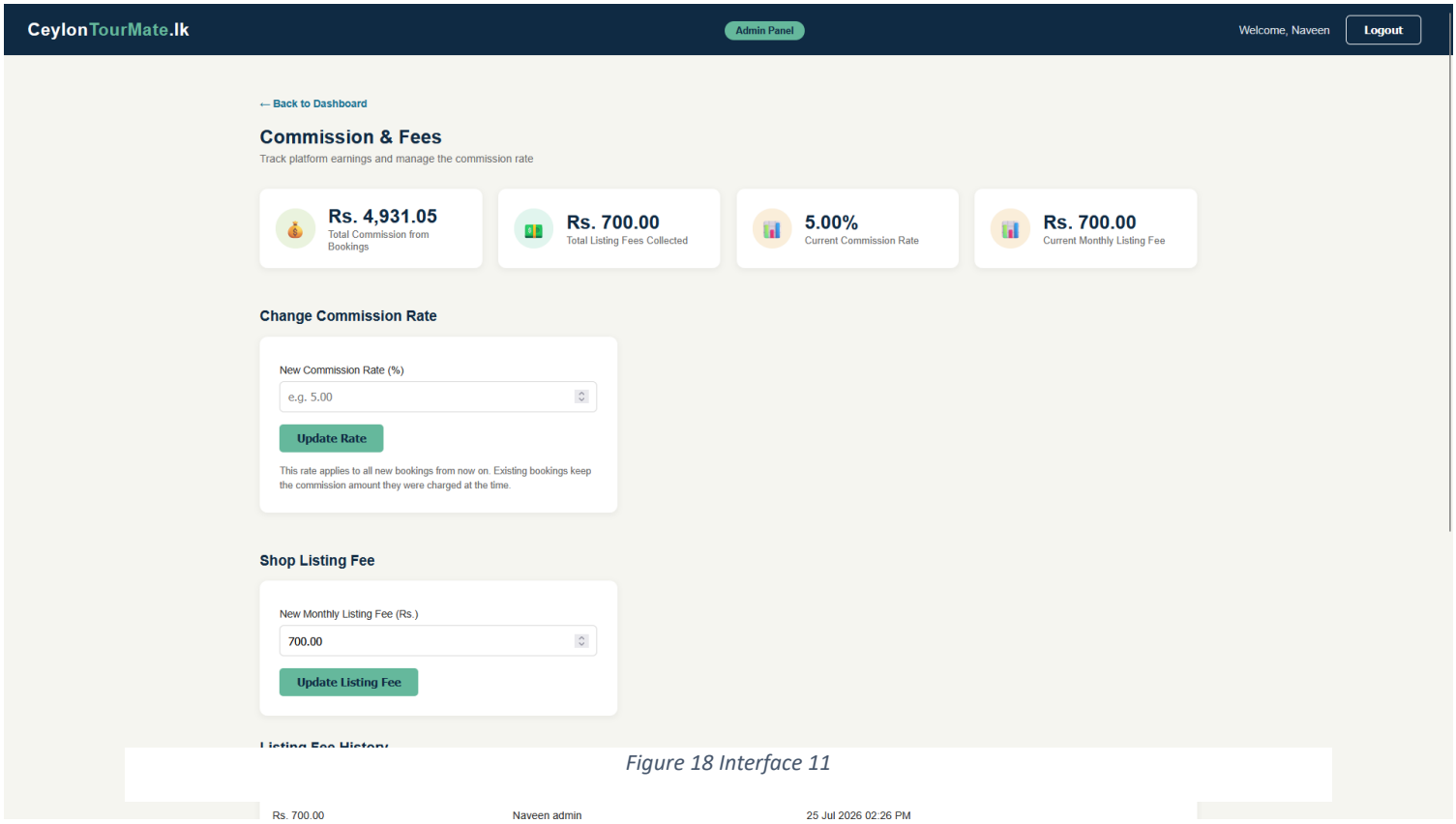


Figure 18 Interface 11

- INT -12 Send notification – Admin can send notifications to users

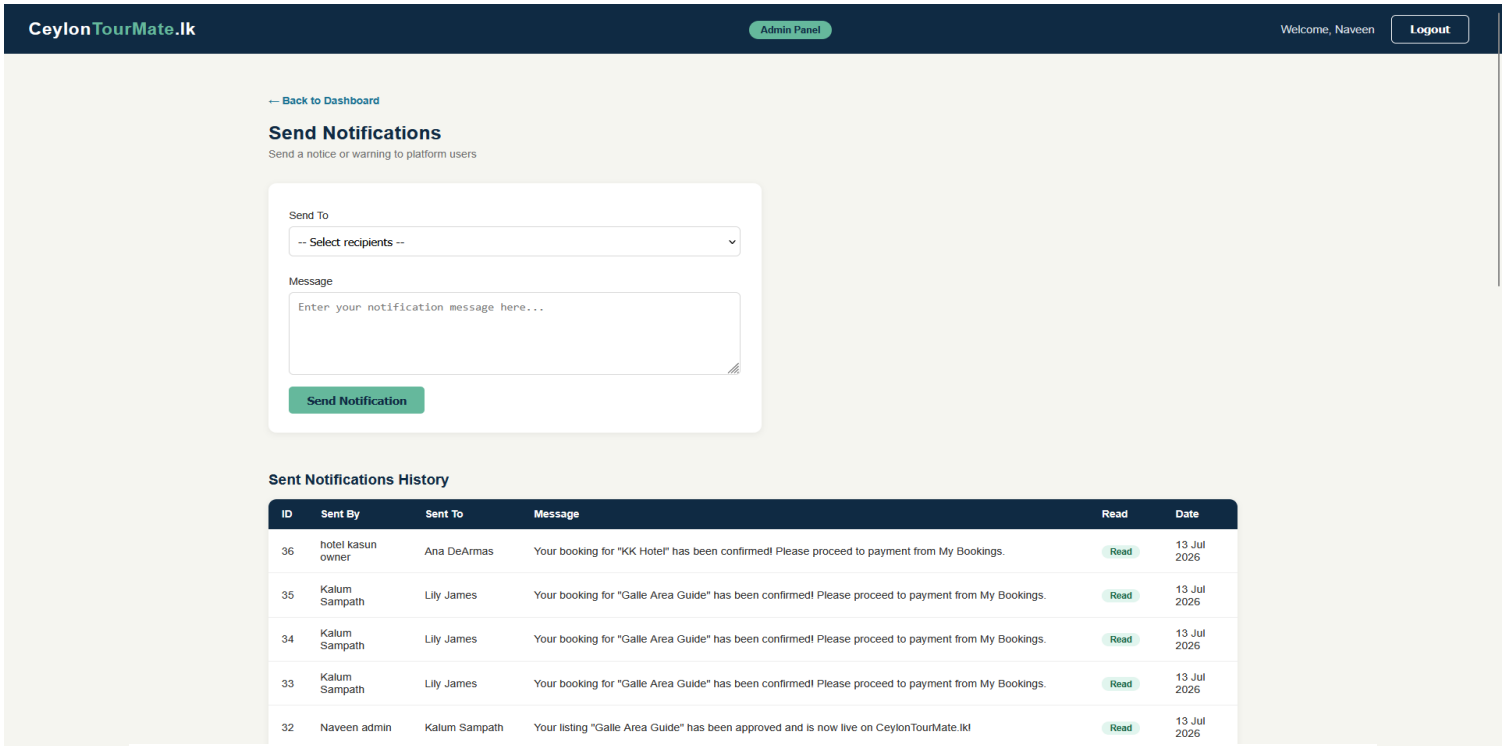


Figure 19 Interface 12

- INT -13 My profile – users can change their profile (same interface for everyone)

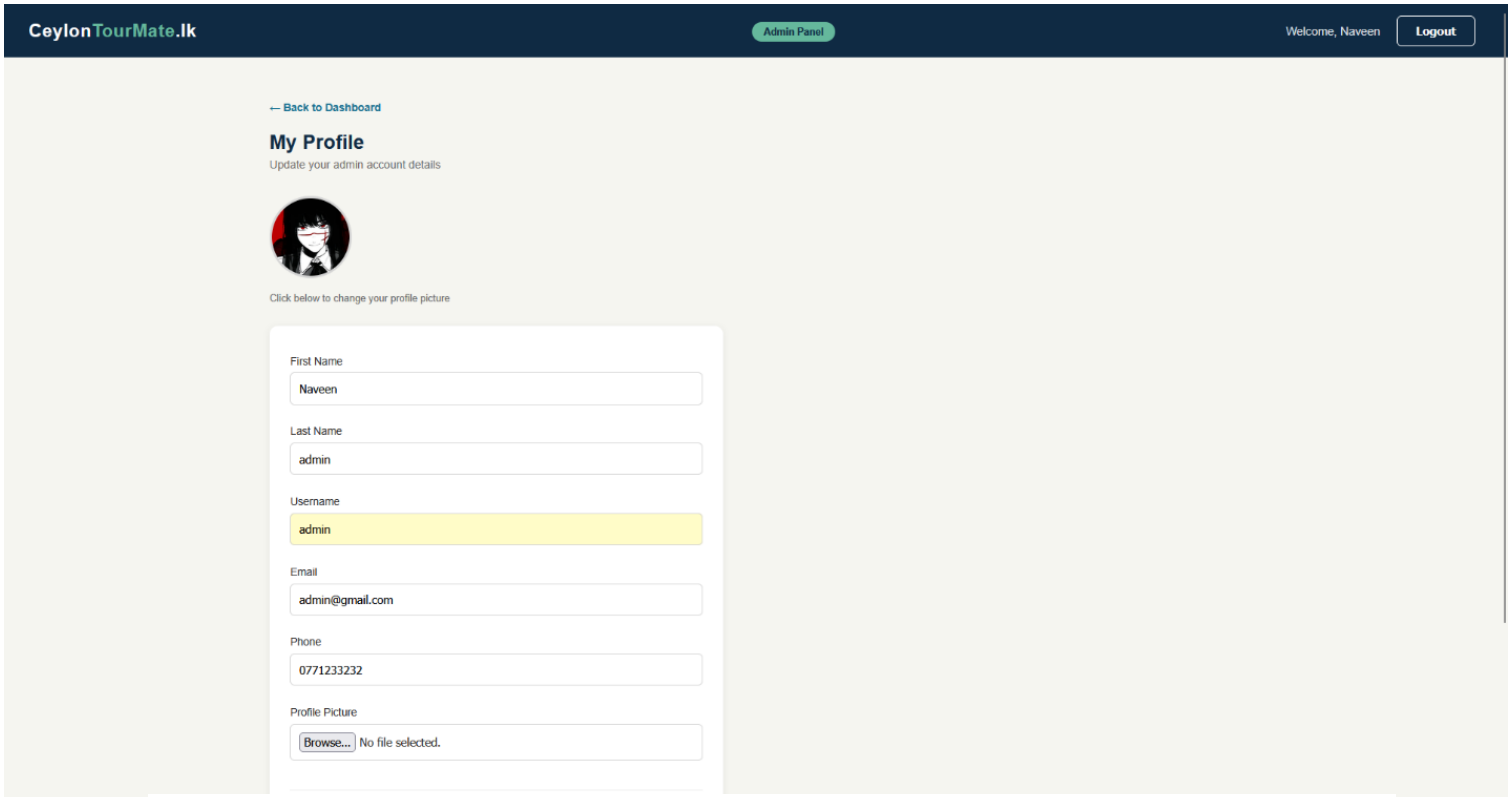


Figure 20 Interface 13

- INT -14 Reports – Admin can generate reports

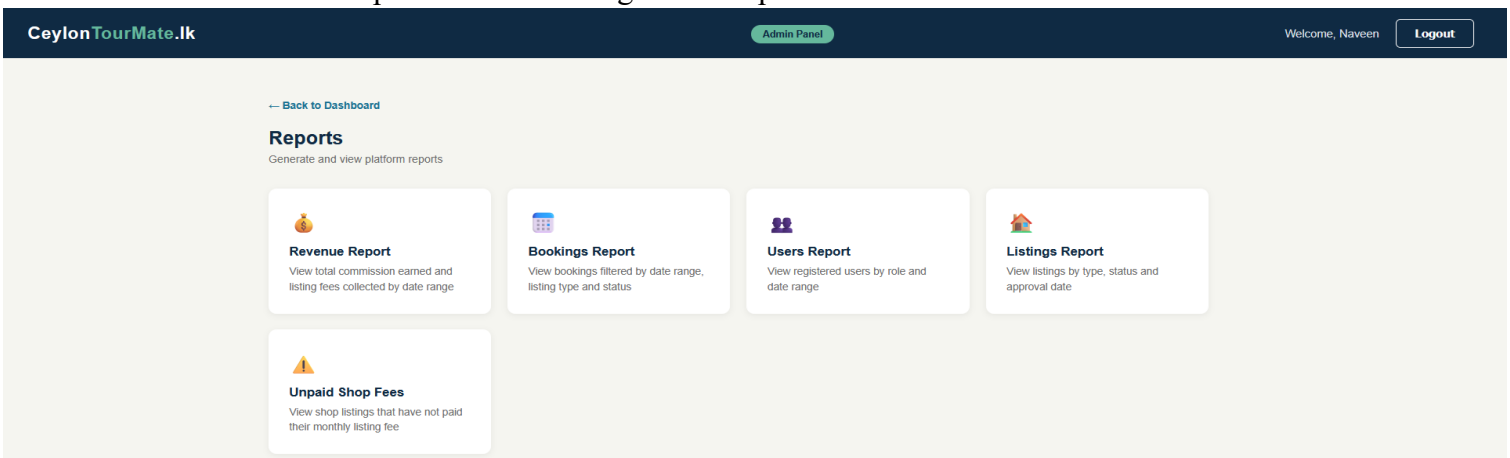


Figure 21 Interface 14

- INT -14 Experience provider dashboard

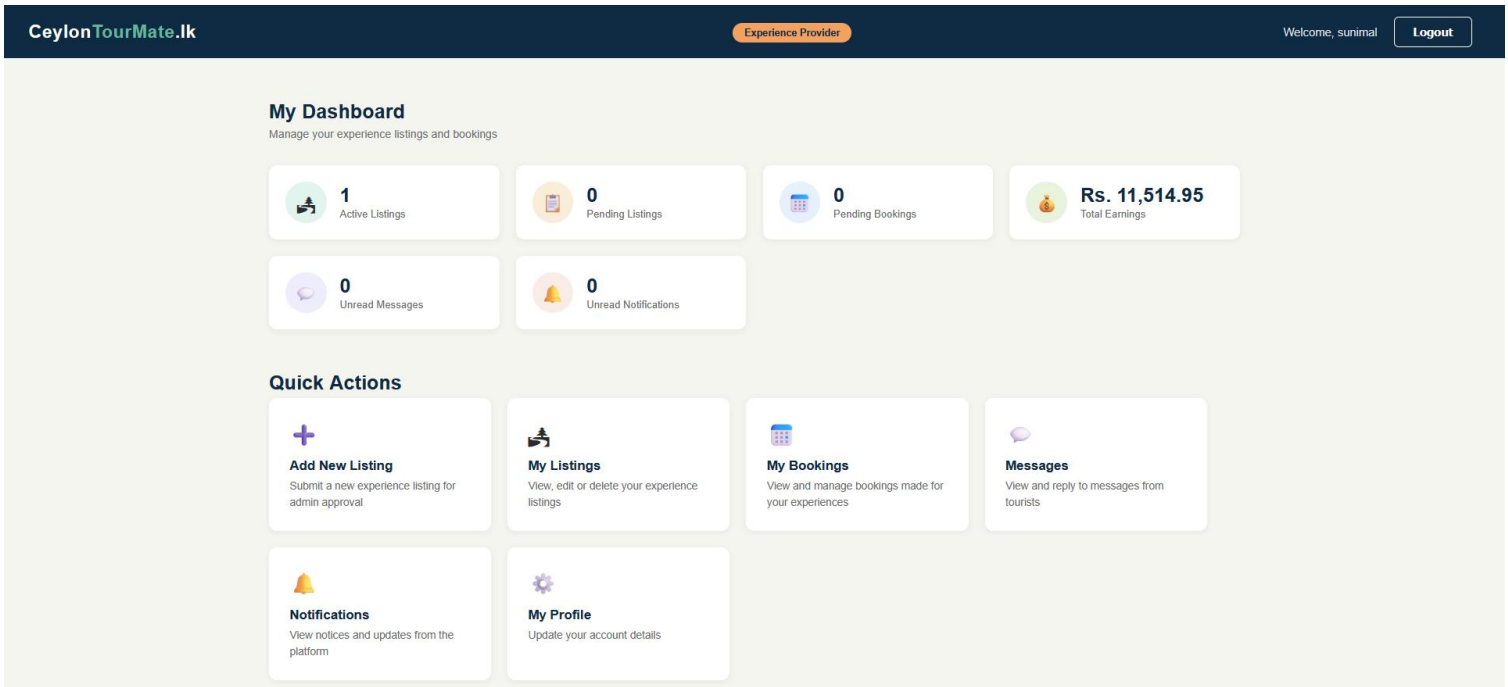


Figure 22 Interface 15

- INT -15 Hotel owner dashboard

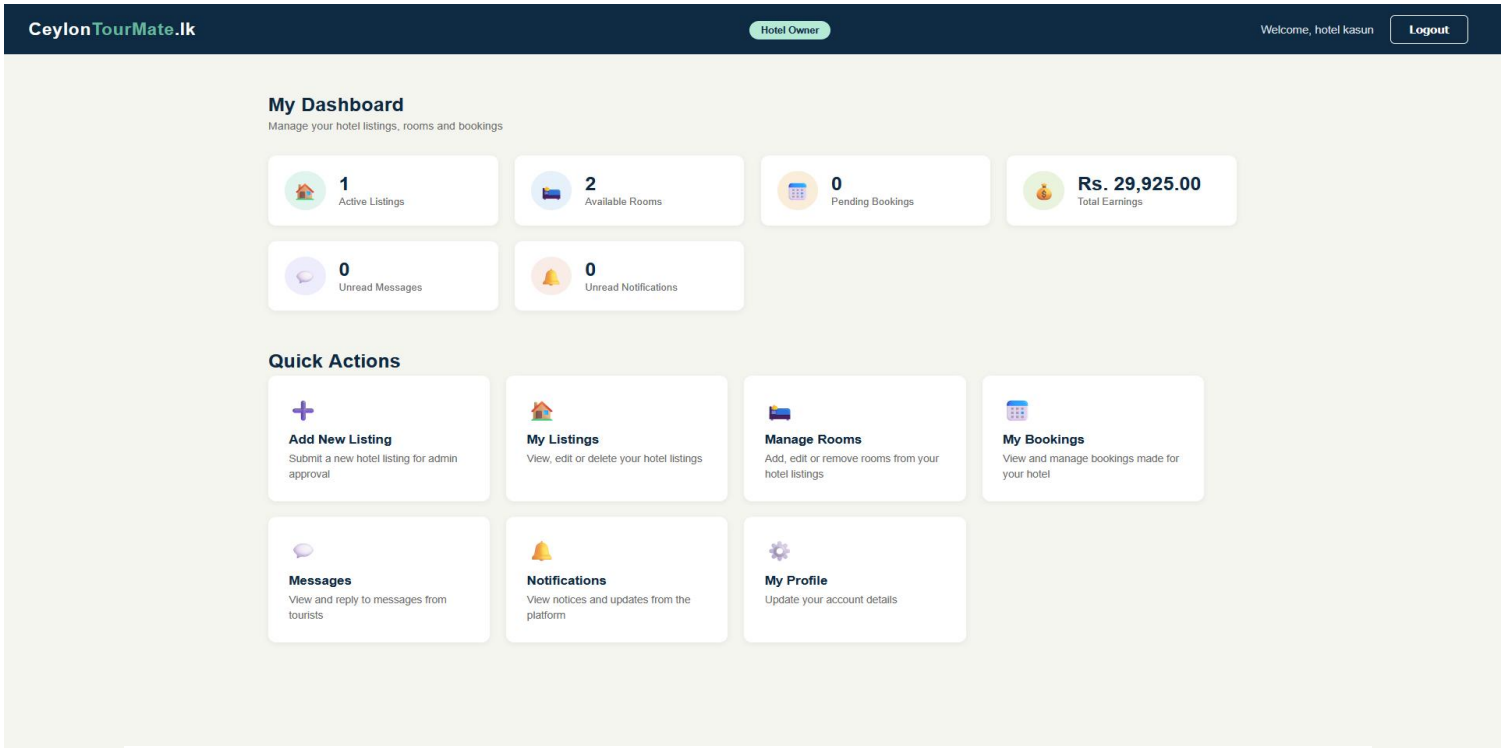


Figure 23 Interface 15

- INT -16 Shop owner dashboard

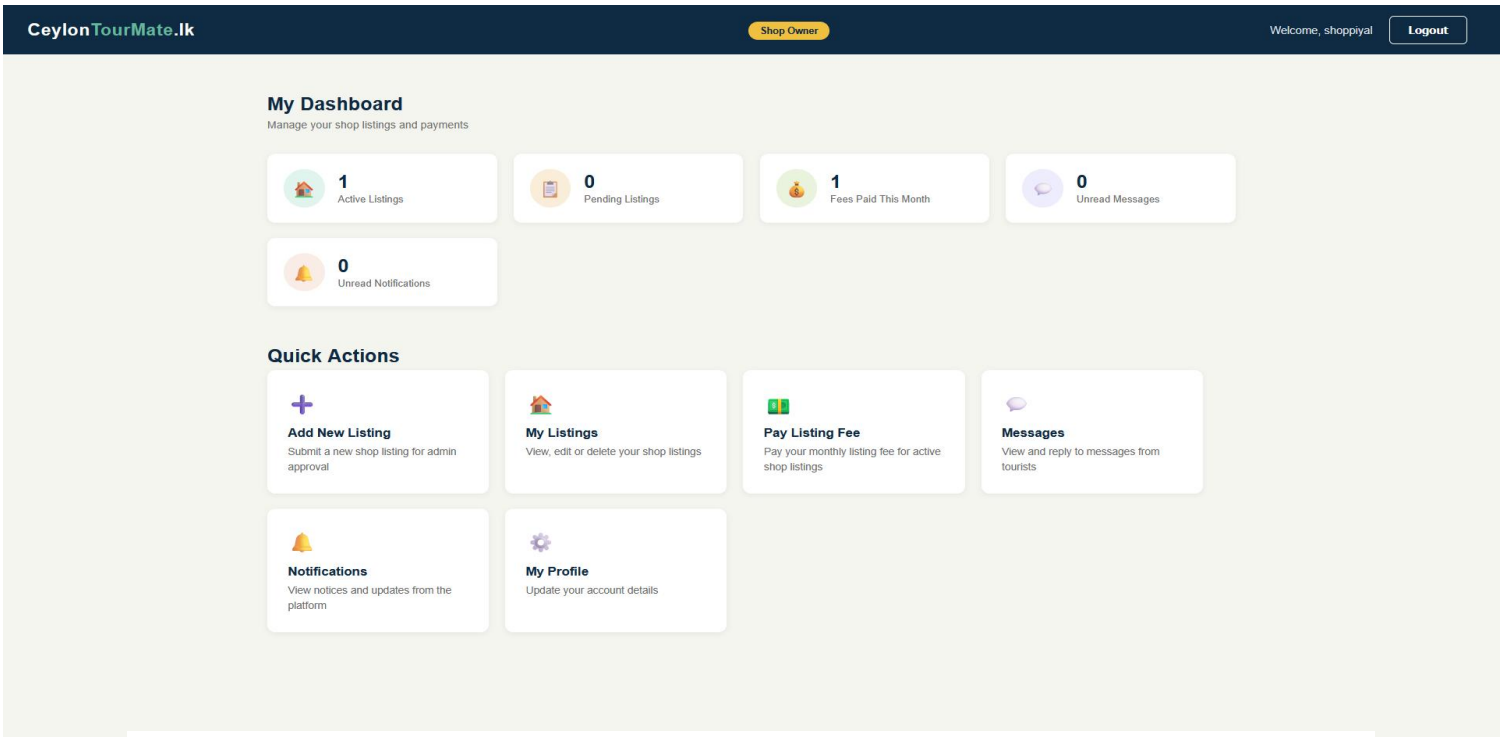


Figure 24 Interface 16

- INT -17 Tour guide dashboard

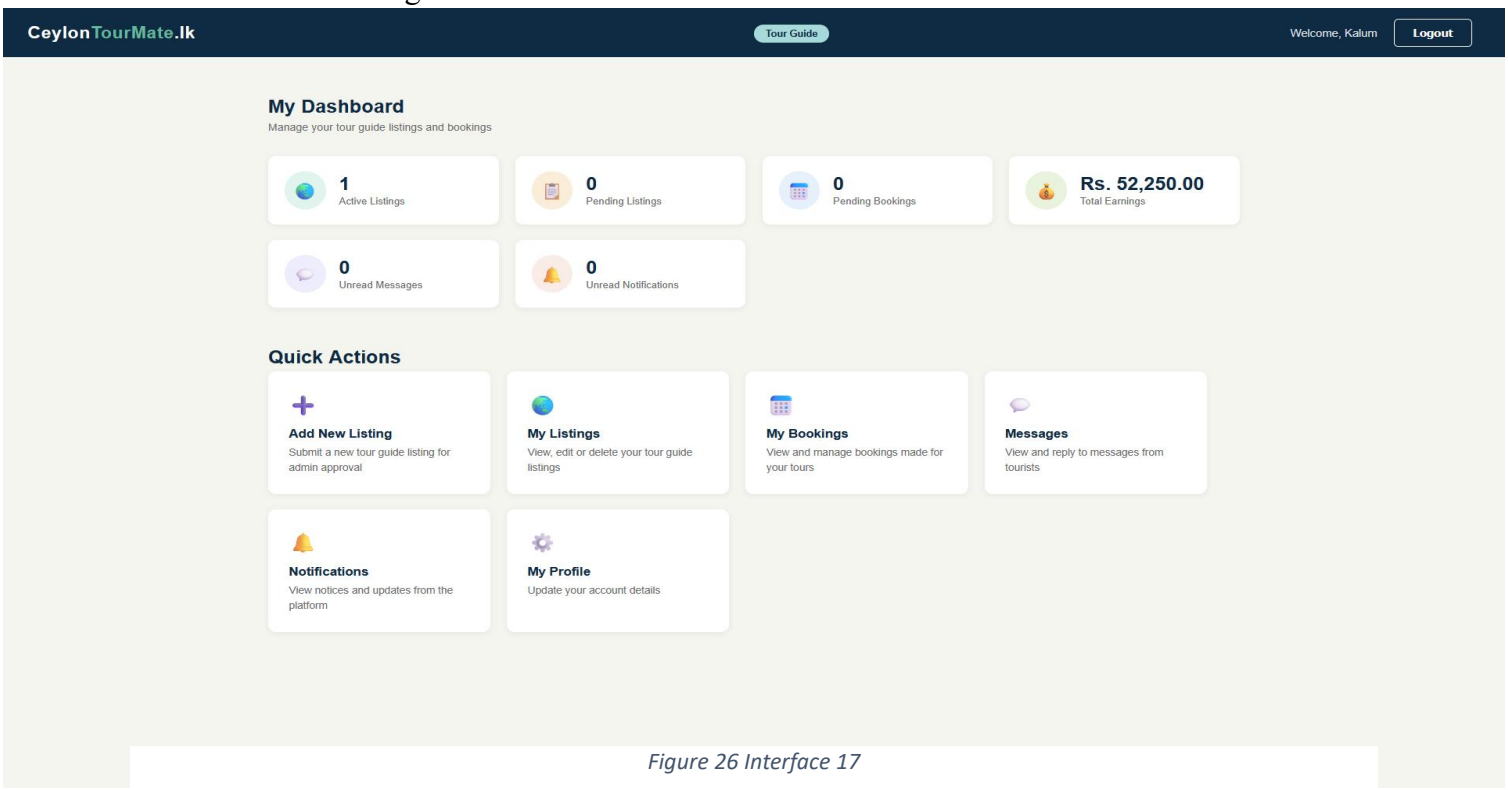


Figure 26 Interface 17

- INT -18 Tourist dashboard

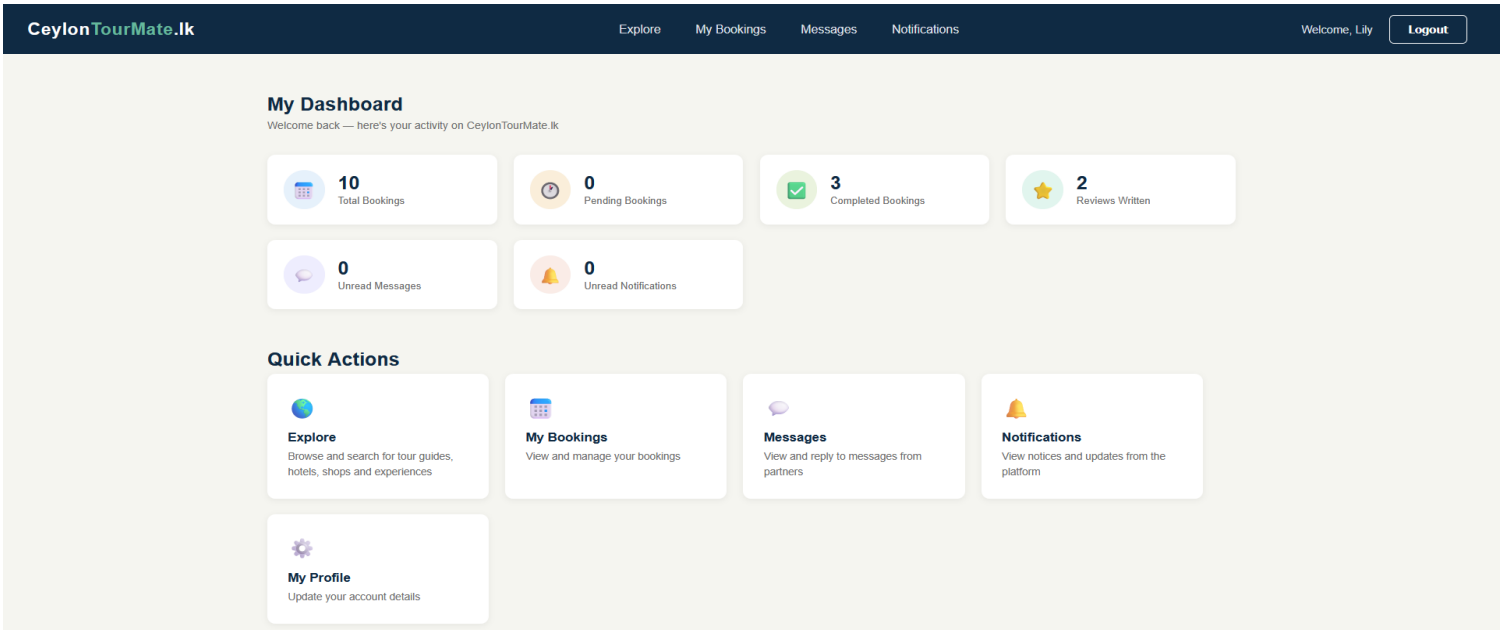


Figure 25 Interface 18

- INT -19 Explorer page – Tourist can explore services

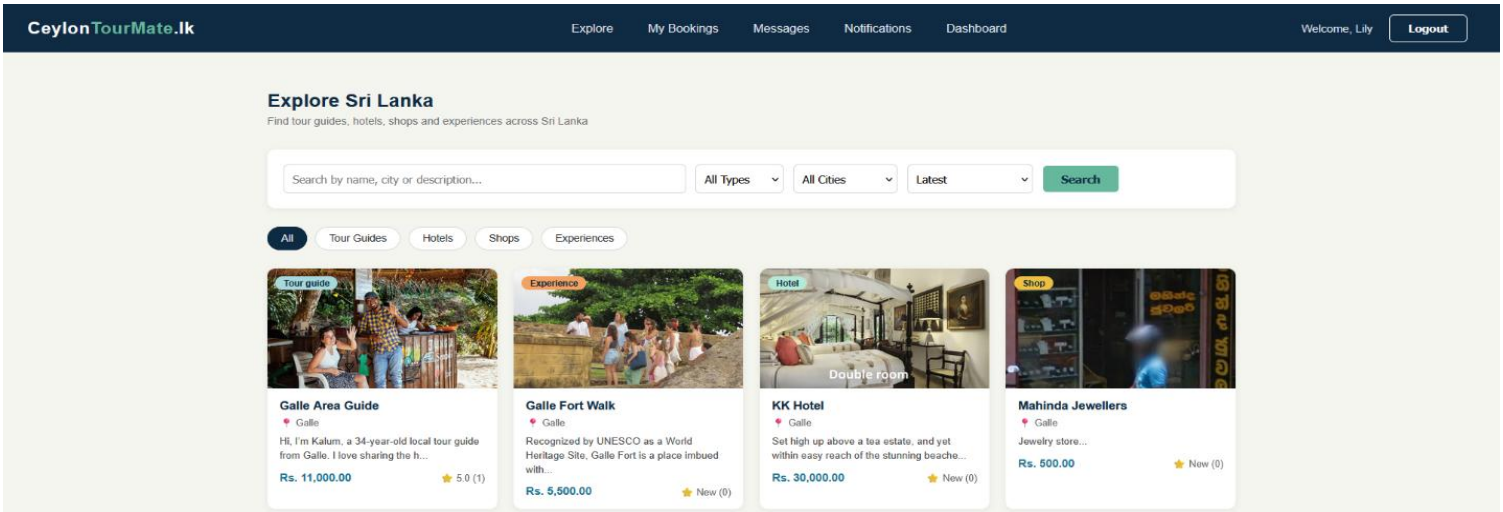


Figure 27 Interface 19

- INT -20 Tourist booking page

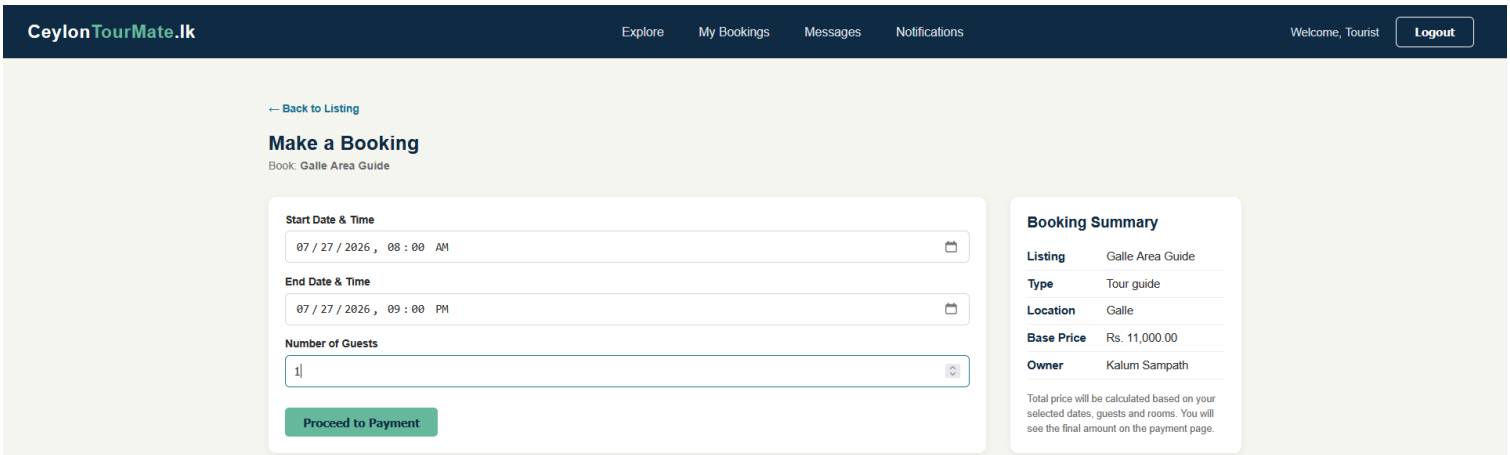


Figure 28 Interface 20

4.3 Database Design

Table: users

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each user (Primary Key)
firstname	Text	50	User's first name
lastname	Text	50	User's last name
username	Text	50	Must be unique
email	Text	100	Must be unique
password	Text	255	Encrypted password
phone	Text	15	Contact number
profile_pic	Text	255	File path to profile picture (default: default.png)
role	Text (fixed options)	1	Options: tourist, tour_guide, shop_owner, hotel_owner, experience_provider, admin
is_suspended	Yes/No	1	Whether the account is suspended by admin
is_approved	Yes/No	1	Whether the account has been approved
security_question	Text	255	Used for password recovery
security_answer	Text	255	Answer to security question
created_at	DateTime	10	Date and time account was created

Table3:Users

Table: listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each listing (Primary Key)
user_id	Number	4	Foreign Key → links to Users table (owner of listing)
title	Text	255	Listing title
description	Text (Long)	100	Full description of the listing
location	Text	255	Specific address/area
city	Text	100	City name
price	Currency	5	Base price of listing
status	Text (fixed options)	1	Options: pending, approved, rejected
type	Text (fixed options)	1	Options: tour_guide, shop, hotel, experience
created_at	DateTime	10	Date and time listing was created
is_deleted	Yes/No	1	Soft-delete flag (1 = hidden, not permanently removed)

Table4:listings

Table: listing_images

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each image (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
image_path	Text	255	File path to image, not the image itself

Table5:listing_images

Table: hotel_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
star_rating	Number	4	Hotel rating from 1 to 5 stars
amenities	Text (Long)	100	Comma-separated list of hotel amenities
total_rooms	Number	4	Total number of rooms in the hotel
check_in_time	DateTime (Time only)	10	Standard check-in time
check_out_time	DateTime (Time only)	10	Standard check-out time

Table6:hotel_listings

Table: rooms

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
hotel_listing_id	Number	4	Foreign Key → links to Hotel Listings table
description	Text (Long)	100	Room description
room_type	Text (fixed options)	1	Options: single, double, suite, family
price_per_night	Currency	5	Price for one night
capacity	Number	4	Maximum number of guests
is_available	Yes/No	1	Whether the room is currently available

Table7:rooms

Table: shop_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
shop_type	Text	100	e.g. Jewellery, Souvenir
open_time	DateTime (Time only)	10	Shop opening time
close_time	DateTime (Time only)	10	Shop closing time

Table8:shop_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
experience_type	Text	100	e.g. Historical, Photography
duration	Text	100	e.g. "2 hours"
max_group_size	Number	4	Maximum people allowed per group
availability	Text	255	e.g. "Daily", "Weekends only"

Table9:experience_listings

Table: tourguide_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
languages	Text	255	Languages spoken by the guide
specialization	Text	255	e.g. Historical, Adventure
experience_years	Number	4	Years of guiding experience
availability	Text	255	e.g. "Daily", "Weekdays only"

Table10:tourguide_listings

Table: tourguide_licenses

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
user_id	Number	4	Foreign Key → links to Users table
license_number	Text	100	Official tour guide license number
license_image	Text	255	File path to uploaded license image

Table11:tourguide_licenses

Table: bookings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
tourist_id	Number	4	Foreign Key → links to Users table (tourist who booked)
listing_id	Number	4	Foreign Key → links to Listings table
status	Text (fixed options)	1	Options: pending, confirmed, cancelled, completed
start_datetime	DateTime	10	Booking start date and time
end_datetime	DateTime	10	Booking end date and time
guest_count	Number	4	Number of guests included in booking
total_price	Currency	5	Total price of the booking
commission_amount	Currency	5	Platform's commission earned from this booking
created_at	DateTime	10	Date and time booking was made

Table12:bookings

Table: booking_rooms

Field Name	Data Type	Size (bytes)	Remarks
booking_id	Number	4	Foreign Key → links to Bookings table (part of Composite Primary Key)
room_id	Number	4	Foreign Key → links to Rooms table (part of Composite Primary Key)

Table13:booking_rooms

Table: payments

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
user_id	Number	4	Foreign Key → links to Users table
shop_listing_id	Number	4	Foreign Key → links to Shop Listings table (used for listing fee payments)
booking_id	Number	4	Foreign Key → links to Bookings table (used for booking payments)
type	Text (fixed options)	1	Options: booking, listing_fee
amount	Currency	5	Payment amount
payment_status	Text (fixed options)	1	Options: pending, completed, failed
fee_month	Number	4	Month number, used only for listing fee payments
fee_year	Number	4	Year, used only for listing fee payments
payment_date	DateTime	10	Date and time payment was made

Table14:payments

Table: reviews

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
tourist_id	Number	4	Foreign Key → links to Users table
listing_id	Number	4	Foreign Key → links to Listings table
booking_id	Number	4	Foreign Key → links to Bookings table
rating	Number	4	Rating score, restricted to 1–5
comment	Text (Long)	100	Reviewer's written comment
created_at	DateTime	10	Date and time review was posted

Table15:reviews

Table: messages

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
sender_id	Number	4	Foreign Key → links to Users table
receiver_id	Number	4	Foreign Key → links to Users table
listing_id	Number	4	Foreign Key → links to Listings table (message relates to this listing)
message	Text (Long)	100	Message content
is_read	Yes/No	1	Whether the message has been read
created_at	DateTime	10	Date and time message was sent

Table16:messages

Table: notifications

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
sender_id	Number	4	Foreign Key → links to Users table (usually admin/system)
receiver_id	Number	4	Foreign Key → links to Users table
message	Text (Long)	100	Notification content
is_read	Yes/No	1	Whether the notification has been read
created_at	DateTime	10	Date and time notification was created

Table17:notification

Table: commission

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
rate	Currency	3	Commission rate value (percentage or fixed fee)
type	Text (fixed options)	1	Options: booking_commission, listing_fee
updated_by	Number	4	Foreign Key → links to Users table (admin who made the change)
updated_at	DateTime	10	Date and time rate was last updated

Table18:commission

4.4 Report Layout Design

- RL-01 Shop listing report fee report – admin can view shop listings that have not paid the listing fee.



Figure 29 report layout 1

- RL-02 All listings report- admin can view all listing by type

CeylonTourMate.lk

All Listings Separated by Type

Generated on: 25 Jul 2026 08:46 PM

Listing Type	Total	Approved	Pending	Rejected	Avg Rating	Total Reviews
Tour guide	1	1	0	0	5.0	1
Shop	1	1	0	0	N/A	0
Hotel	1	1	0	0	N/A	0
Experience	1	1	0	0	N/A	0

Listings Summary

Total Listings on Platform 4

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 30 report layout 2

- RL-03 All users reports – admin can view all user count on the platform by role

CeylonTourMate.lk

All Time User Statistics by Role

Generated on: 25 Jul 2026 08:46 PM

Role	Total	Active	Pending Approval	Suspended
Tourist	3	3	0	0
Tour guide	2	1	1	0
Shop owner	3	3	0	0
Hotel owner	1	1	0	0
Experience provider	1	1	0	0

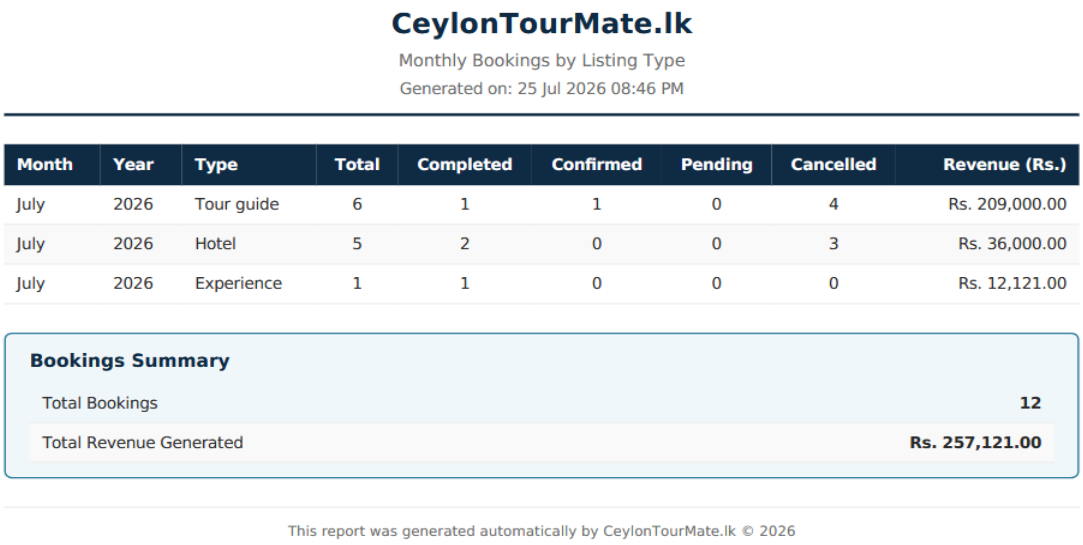
Users Summary

Total Registered Users 10

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 31 report layout 3

- RL-04 Monthly bookings report- admin can view all monthly bookings



CeylonTourMate.lk

Monthly Bookings by Listing Type

Generated on: 25 Jul 2026 08:46 PM

Month	Year	Type	Total	Completed	Confirmed	Pending	Cancelled	Revenue (Rs.)
July	2026	Tour guide	6	1	1	0	4	Rs. 209,000.00
July	2026	Hotel	5	2	0	0	3	Rs. 36,000.00
July	2026	Experience	1	1	0	0	0	Rs. 12,121.00

Bookings Summary

Total Bookings	**12**
Total Revenue Generated	**Rs. 257,121.00**

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 32 report layout 4

- RL-05 Revenue report – admin can view the total platform's revenue.



CeylonTourMate.lk

Monthly Revenue — Commission & Listing Fees

Generated on: 25 Jul 2026 08:46 PM

Booking Commission Revenue

Month	Year	Completed Bookings	Commission Earned (Rs.)
July	2026	4	Rs. 4,931.05

Listing Fee Revenue

Month	Year	Payments Received	Fees Collected (Rs.)
July	2026	1	Rs. 700.00

Revenue Summary

Total Commission from Bookings	**Rs. 4,931.05**
Total Listing Fees Collected	**Rs. 700.00**
Grand Total Revenue	Rs. 5,631.05

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 33 report layout 5

Chapter 5: Implementation

5.1 Introduction

In this chapter im going to explain the main implementation of my project, which covers system architecture, main classes and decisions I made during the coding part.

5.2 System architecture

As the architecture I used **3 tier architecture** for the development of CeylonTourMateLK with OOP phps and MYSQL database.

My main classes are as follows,

Booking.php	2026-07-15 8:21 PM	PHP Source File	13 KB
Commission.php	2026-07-15 8:31 PM	PHP Source File	5 KB
DBconnection.php	2026-07-15 8:28 PM	PHP Source File	1 KB
ExperienceListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
HotelListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
Listing.php	2026-07-15 8:28 PM	PHP Source File	10 KB
ListingImage.php	2026-07-15 8:28 PM	PHP Source File	2 KB
Message.php	2026-07-15 8:28 PM	PHP Source File	5 KB
Notification.php	2026-07-15 8:28 PM	PHP Source File	4 KB
Payment.php	2026-07-15 8:28 PM	PHP Source File	7 KB
Review.php	2026-07-15 8:28 PM	PHP Source File	4 KB
Room.php	2026-07-15 8:28 PM	PHP Source File	5 KB
session.php	2026-07-15 8:28 PM	PHP Source File	2 KB
ShopListing.php	2026-07-15 8:28 PM	PHP Source File	4 KB
TourGuideLicense.php	2026-07-15 8:28 PM	PHP Source File	2 KB
TourGuideListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
User.php	2026-07-15 8:28 PM	PHP Source File	12 KB

Figure 34 Main classes

5.3 Main Coding Areas

I used a session class to hold logged in users session information, which I use it to show a welcome message at the top of the platform.

```php
function checkSession() {
    if (session_status() == PHP_SESSION_NONE) {
        session_start();
    }
    if (!isset($_SESSION['user_id']) || !isset($_SESSION['role'])) {
        header("Location:login.php");
        exit();
    }
    if (!isset($_SESSION['firstname'])) {
        $conn = DBconnection::getInstance()->getConnection();
        $stmt = $conn->prepare("SELECT firstname FROM users WHERE id = ?");
        $stmt->bind_param("i", $_SESSION['user_id']);
        $stmt->execute();
        $result = $stmt->get_result();
        $user = $result->fetch_assoc();
        if ($user) {
            $_SESSION['firstname'] = $user['firstname'];
        }
    }
}
```

Figure 35 Session class

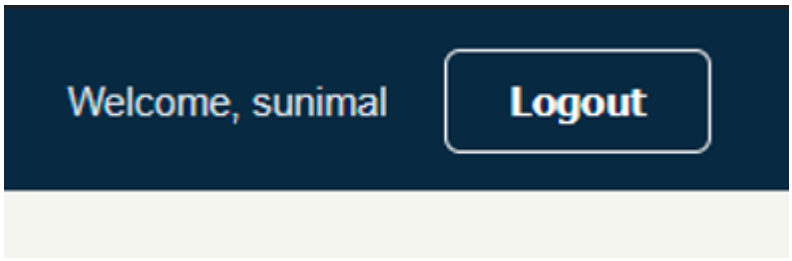


Figure 36 Welcome message

And for my whole project this is my folder structure,

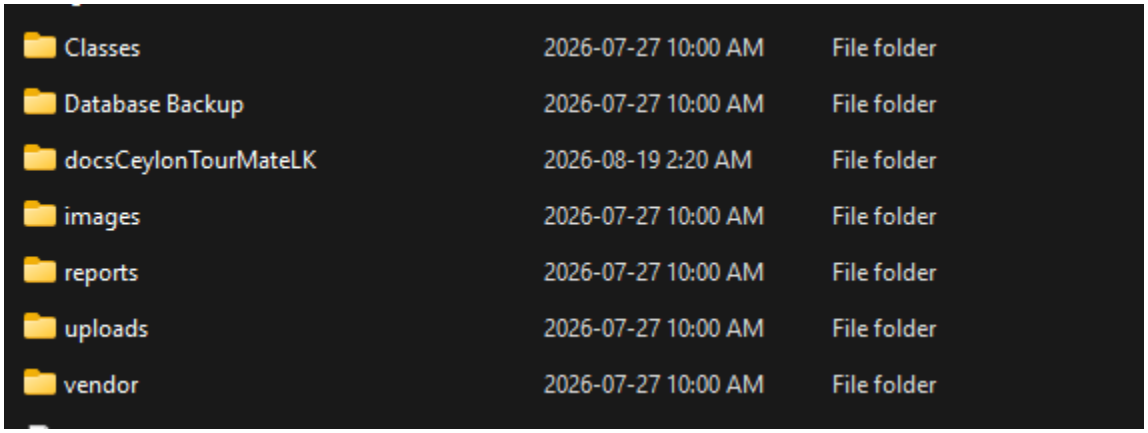


Figure 37folder structure

This is how I coded to connect the database to the system,

```php
<?php

class DBconnection {

    private static $instance = null;
    private $connection;

    private $host     = "localhost";
    private $username = "root";
    private $password = "";
    private $database = "ceylontourmatelk";

    private function __construct() {
        $this->connection = new mysqli($this->host, $this->username, $this->password, $this->database);

        if ($this->connection->connect_error) {
            throw new Exception("Database connection failed: " . $this->connection->connect_error);
        }

        $this->connection->set_charset("utf8mb4");
    }

    public static function getInstance() {
        if (self::$instance == null) {
            self::$instance = new DBconnection();
        }
        return self::$instance;
    }

    public function getConnection() {
        return $this->connection;
    }

}
?>
```

Figure 38 DB code

For extra security I used IF ELSE statements to show error messages,

```
<div class="auth-section">
    <div class="auth-card auth-card-narrow">
        <h2>Welcome back</h2>
        <p class="auth-subtitle">Login to your CeylonTourMate.lk account</p>

        <?php if(isset($_GET['registered'])) { ?>
            <p class="success">Registration successful! You can now log in.</p>
        <?php } ?>
        <?php if(isset($_GET['pending'])) { ?>
            <p class="success">Registration submitted! Please wait for admin approval before logging in.</p>
        <?php } ?>
        <?php if(isset($_GET['wrongpassword'])) { ?>
            <p class="error">Incorrect password. Please try again.</p>
        <?php } ?>
        <?php if(isset($_GET['notfound'])) { ?>
            <p class="error">No account found with that username or email.</p>
        <?php } ?>
        <?php if(isset($_GET['suspended'])) { ?>
            <p class="error">Your account has been suspended. Please contact admin.</p>
        <?php } ?>
        <?php if(isset($_GET['notapproved'])) { ?>
            <p class="error">Your account is pending admin approval.</p>
        <?php } ?>
        <?php if(isset($_GET['error'])) { ?>
            <p class="error">Something went wrong. Please try again.</p>
        <?php } ?>
        <?php if(isset($_GET['resetdone'])) { ?>
            <p class="success">Password reset successful! You can now log in.</p>
        <?php } ?>
        <?php if(isset($_GET['loggedout'])) { ?>
            <p class="success">You have been logged out successfully.</p>
        <?php } ?>
        <form name="loginform" method="post" action="loginbackend.php" onsubmit="return validateLoginForm();">
            <label>Username or Email</label>
```

Figure 39 if else code

To avoid sql injections and for the security I used prepared statements where necessary,

```php
public function register() {
    $conn = DBconnection::getInstance()->getConnection();

    $checkSql = "SELECT id FROM users WHERE username = ? OR email = ?";
    $checkStmt = $conn->prepare($checkSql);
    $checkStmt->bind_param("ss", $this->username, $this->email);
    $checkStmt->execute();
    $checkStmt->store_result();

    if ($checkStmt->num_rows > 0) {
        return "exists";
    }

    $hashedPassword = password_hash($this->password, PASSWORD_DEFAULT);

    $sql = "INSERT INTO users (firstname, lastname, username, email, password, phone,
            profile_pic, role, is_suspended, is_approved, security_question, security_answer)
            VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?)";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param("sssssssiiss",
        $this->firstname,
        $this->lastname,
        $this->username,
        $this->email,
        $hashedPassword,
        $this->phone,
        $this->profilePic,
        $this->role,
        $this->isSuspended,
        $this->isApproved,
        $this->securityQuestion,
        $this->securityAnswer
    );
```

Figure 40 prepared statements

Using Mpdf I generated reports by coding sql queries as follows,

```php
<?php
if (session_status() == PHP_SESSION_NONE) {
    session_start();
}
require_once '../Classes/session.php';
checkRole(['admin']);

require_once '../Classes/DBconnection.php';
require_once 'pdfhelper.php';

$conn = DBconnection::getInstance()->getConnection();

$sql  = "SELECT
            MONTHNAME(b.created_at) as month_name,
            YEAR(b.created_at) as year,
            l.type as listing_type,
            COUNT(b.id) as total_bookings,
            SUM(CASE WHEN b.status = 'completed' THEN 1 ELSE 0 END) as completed,
            SUM(CASE WHEN b.status = 'confirmed' THEN 1 ELSE 0 END) as confirmed,
            SUM(CASE WHEN b.status = 'pending' THEN 1 ELSE 0 END) as pending,
            SUM(CASE WHEN b.status = 'cancelled' THEN 1 ELSE 0 END) as cancelled,
            COALESCE(SUM(b.total_price), 0) as total_revenue
        FROM bookings b
        JOIN listings l ON b.listing_id = l.id
        GROUP BY YEAR(b.created_at), MONTH(b.created_at), l.type
        ORDER BY YEAR(b.created_at) DESC, MONTH(b.created_at) DESC, l.type";
$stmt = $conn->prepare($sql);
$stmt->execute();
$bookings = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);

$totalBookings = array_sum(array_column($bookings, 'total_bookings'));
$totalRevenue  = array_sum(array_column($bookings, 'total_revenue'));

$mpdf = createPDF('Bookings Report');
$html = getReportStyles();
$html .= getReportHeader('Bookings Report', 'Monthly Bookings by Listing Type');
```

Figure 41 mPDF code

Since the beginning I use github as my version control platform and here are some insights,

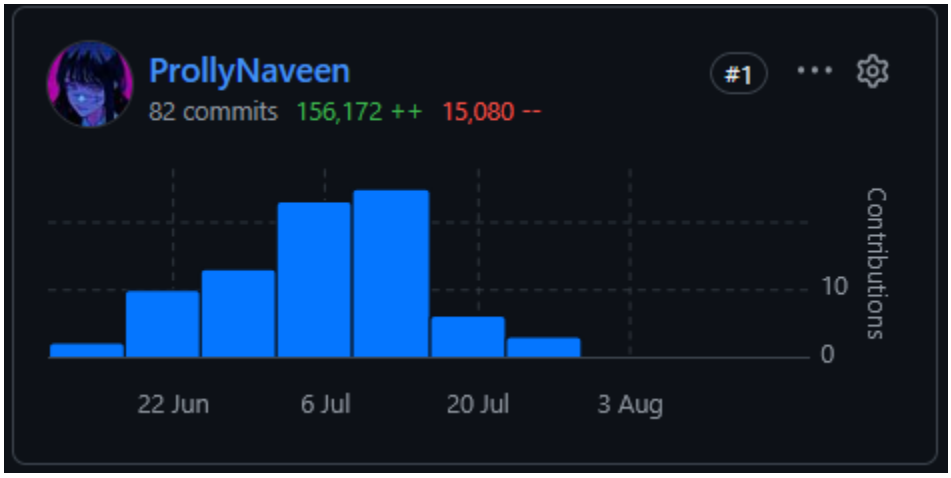


Figure 42githubcommits

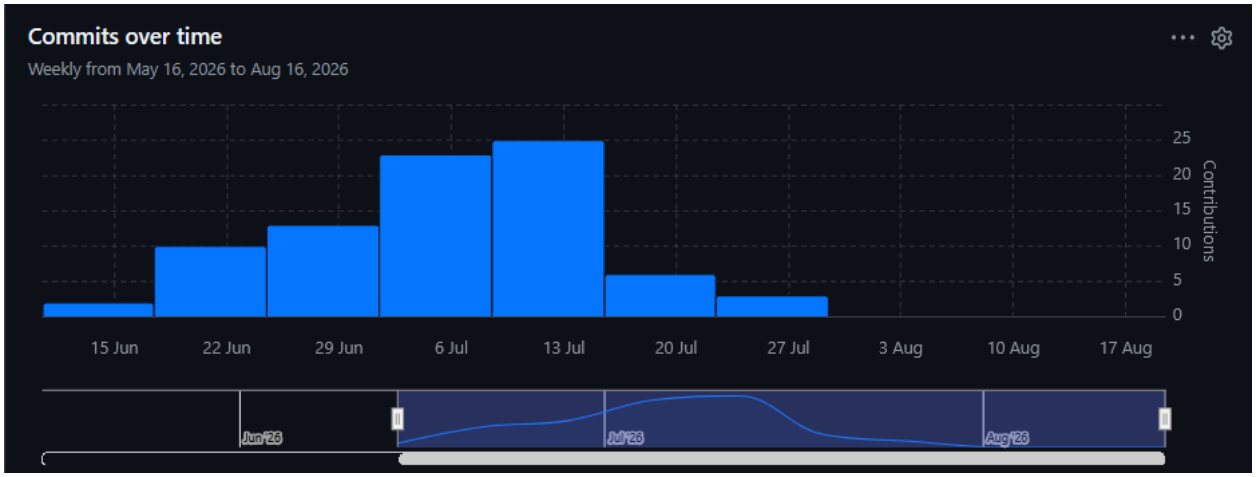


Figure 43commits over time

5.4 Admin Report Implementation

For admins report function I used the mPDF library which was easy to code and it generates pdfs using data directly from the database.



Figure 44 mPDF

Chapter 6: Testing

These are some test cases I used to test and confirm the functionalities of my system.

TEST ID	TEST CASE	INPUT	EXPECTED RESULT	ACTUAL RESULT
TC-01	**Tourist registration with valid data**	Valid email, username, password	Account created successfully	**Pass**
TC-02	**Tourist registration with duplicate email**	Existing email address	Error: account already exists	**Pass**
TC-03	**Login with correct credentials**	Valid username/password	Redirected to role-specific dashboard	**Pass**
TC-04	**Login with incorrect password**	Valid username, wrong password	Error: wrong password	**Pass**
TC-05	**Partner registration pending approval**	New tour guide registration	Account created with `is_approved = 0`, login blocked until approved	**Pass**
TC-06	**Add a new shop listing**	Valid listing form data	Listing created with status "pending"	**Pass**
TC-07	**Edit listing — readonly fields protected**	Attempt to submit form with unchanged readonly fields	Title/location/city unchanged, only description/price/hours updated	**Pass**
TC-08	**Delete a listing**	Click delete on an owned listing	Listing hidden from all views (`is_deleted = 1`), underlying data preserved	**Pass**
TC-09	**Tour guide booking price calculation**	3-day booking, price Rs. 11,000/day	Total price = Rs. 33,000	**Pass**
TC-10	**Hotel booking with multiple rooms**	2 rooms selected, 3 nights	Total price = sum of (room price × nights) for both rooms	**Pass**
TC-11	**Booking with unavailable room**	Room marked unavailable selected	Error: room unavailable	**Pass**
TC-12	**Confirm booking (partner)**	Partner confirms a pending booking	Status updated to "confirmed", notification sent to tourist	**Pass**
TC-13	**Attempt to edit another user's listing**	Listing ID belonging to a different user	Error: unauthorized	**Pass**
TC-14	**Send message to partner**	Tourist sends message from listing page	Message saved, visible in both users' message threads	**Pass**
TC-15	**Write review after completed booking**	Tourist submits review for completed booking	Review saved and displayed on listing page	**Pass**

Table19Test cases

Chapter 7: Conclusion

7.1 Achievements

As a conclusion I would like to describe what I achieved from my project, CeylonTourMateLK.

- A fully functional centralized platform connecting all tourists and local service providers
- Role based login to make the system easy to use for users
- A complete way for tourists to go form searching booking confirmation to payments.
- System sends automatic notifications to users(ex: when local partner approves a booking tourist receives a notification)
- Throughout the project I have learned all the knowledge of SDLC process specially with the diagrams and coding part.

7.2 Limitations

- The main limitations is the payment system still implement a payment processing rather than a real payment gateway.
- I need to find users for this platform by advertising.

7.3 Future Developments

- My main goal is to find users and give them the knowledge to how to use this platform and advantages of this one compared to the existing third party sites.
- In the future I would like to integrate this with a real payment gateway for live transactions.
- I would like to build a dedicated mobile app for tourist to connect with this website too.

7.4 Final Conclusion

CeylonTourMateLK will be a great platform for community as well as my self too. From the beginning I have learned everything again to make this project a success, all the things is a original work done my me, my self.

This platform address a huge gap in the tourism industry in srilanka and as a solution for that provide a centralized trustworthy platform for the tourists.

References

- Stack Overflow. (n.d.). *How to model messages exchanged between users? - ER Diagram*. [online] Available at: https://stackoverflow.com/questions/35310283/how-to-model-messages-exchanged-between-users-er-diagram [Accessed 26 July 2026].
- asdad. (2026). *Reddit - please wait for verification*. Reddit.Com. https://www.reddit.com/r/sql/comments/13ppmpj/what_do_you_think_of_this_er_diagram_i_feel_like/?utm_source=chatgpt.com
- table, I. (2014, November 15). *Stack overflow*. Stack Overflow. https://stackoverflow.com/questions/26950903/is-it-okay-for-the-foreign-key-field-to-have-a-different-name-than-the-field-it

Progress Monitoring Sheet

Final Project

NIBM POWERING GREAT MINDS

Course Title : Diploma in Software Engineering

Student Index : K.K. Naveen Rashmika Devinda
GADSE251F-018

Project Title : Web based tourism services marketplace platform "CeylonTourMate.lk"

Supervisor Name : Ms. Ashanthi KyrakylaSyriya

Meeting Date	Students Indices (participated)	Supervisor Comments	Signature of the supervisor
26/6/8	gadse251f-018	Use case diagram ER diagram	
26/6/18	GADSE251F-018	- ER to be finalized - class diagram to be finalized - Design to be started	
23/6/26	GADSE251F-018	finalize ER and confirm with DB Implementation and functional Design	
23/7/26	GADSE251F-018	complete the report and complete the system functionalities	

Figure 45 progress monitoring sheet